Cross-Study Meta-Analysis of the Gut-Microbiota-Microglia Axis Uncovers Cell-Intrinsic Interferon Shutoff, Invariant Nutrient-Sensing Adapters, and Multi-Omic Reversibility

Samyak Meshram[1,*], **Dr. Soumya Dhokey**[1]

[1]Independent Researchers, Open-Science Initiative

[*]Correspondence: sam.meshram18@gmail.com | GitHub: https://github.com/samyakmeshram/NeuroGut-MetaSeq

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Abstract

Microglia are the resident immune sentinels and parenchymal phagocytes of the central nervous system (CNS), continuously calibrated by biochemical cues from the indigenous gut microbiota. While individual transcriptomic studies have revealed that gut microbiome depletion compromises microglial ramification and immune alertness, findings across laboratories have suffered from small sample sizes, isolation technology carryover, and disparate experimental models. Here, we present **NeuroGut-MetaSeq (v1.2.1)**, an integrated multi-cohort and multi-omic meta-analysis framework synthesizing 60 biological transcriptomes across four independent rodent cohorts spanning lifelong germ-free (GF) isolation, acute broad-spectrum antibiotic (ABX) cocktails, and dietary fiber starvation. Using negative binomial generalized linear models, Restricted Maximum Likelihood (REML) variance estimation, and Hartung-Knapp-Sidik-Jonkman (HKSJ) small-sample random-effects pooling across 23,096 common genes, we separate an invariant multi-study core from model-private perturbation shocks. We identify an omnipresent invariant core led by the basolateral polarity adapter *Llgl2* ($k = 4, \hat{\theta}_{\text{RE}} = +0.672$, HKSJ 95% CI $[0.166, 1.178]$, $I^2 = 0.0\%$) and extracellular chaperone hub *Clu* ($k = 3, \hat{\theta}_{\text{RE}} = +0.850, I^2 = 0.0\%$), alongside universal loss of myeloid quiescence via *Slfn2* ($k = 4, \hat{\theta}_{\text{RE}} = -0.465, p_{\text{HKSJ}} = 0.0197$) and chromatin corepressor *Sap30* ($I^2 = 0.0\%, p_{\text{HKSJ}} = 0.0409$). Paradoxically, acute antibiotic shock markers *Tsc22d3* (GILZ) and *Ddit4* (REDD1) exhibit extreme between-study heterogeneity ($I^2 > 95\%$), classifying them as transient perturbation effectors rather than core dysbiosis hits.

Factorial sex-interaction modeling across 51 sex-informative samples reveals that **99.1% of the microglial response to microbiome depletion is sex-invariant** ($I^2_{\text{sex}} = 0.0\%$), with master regulators *Irf1* ($p = 0.863$) and *Stat1* ($p = 0.362$) showing identical shutdown across sexes, while uncovering selective male-biased vulnerability in quiescence checkpoint *Slfn2* ($\hat{\theta}_{\text{int}} = +0.338, p = 0.017$) and *Oas1a* ($\hat{\theta}_{\text{int}} = +0.335, p = 0.051$). Whole-transcriptome Gene Set Enrichment Analysis (GSEA) and TRRUST upstream regulon deconvolution establish the collapse of basal tonic interferon surveillance (`Hallmark Interferon Gamma Response` NES = -2.392, FDR = 0.0; `Interferon-Responsive Microglia` NES = -2.086), pinpointing master transcription factor **IRF1** as the upstream regulon driver ($Z = -2.284, p = 0.0013$, FDR = 0.0384). High-resolution single-cell deconvolution (BayesPrism, $\kappa = 1.54 < 30$) establishes that Interferon-Responsive Microglia (IRM) are physically preserved (17.5% depleted vs 16.8% reference), proving that the collapse is a genuine **cell-intrinsic per-cell transcriptional shutoff** ($p < 0.001$).

Tripartite microglial ATAC-seq footprinting (Erny 2021) directly demonstrates that TOBIAS open chromatin footprints at *Irf1* and *Stat1* promoter motifs collapse during depletion and are **88.9% restored** upon short-chain fatty acid (SCFA) repletion. In silico NicheNet ligand-receptor prioritization identifies circulating bacterial outer membrane vesicles (TLR4/CD14, $r = 0.658$) and brain microvascular endothelial *Ifnb1* (IFNAR1/2, $r = 0.600$) as primary upstream drivers of basal IRF1 tone. Finally, co-expression analysis reveals coordinate upregulation of *Llgl2* and the large neutral amino acid transporter **LAT1 (*Slc7a5*)** (+0.641 LFC) alongside mTOR suppression (-0.469 LFC), establishing an invariant nutrient-scavenging adaptation. We resolve the in vivo blood-brain barrier (BBB) pharmacokinetic paradox through a Three-Pillar relay framework (border-associated macrophages, ACSS2/acetate central metabolic replenishment, and vagal signaling). Together, these multi-omic findings prove that gut microbiome depletion induces an aberrant hybrid state of blunted antiviral surveillance and metabolic scavenging characterized by **chromatin-poised transcriptional reversibility** upon microbial metabolite restoration.

Significance Statement:

Communication along the gut-brain axis is essential for neurodevelopment and immune homeostasis, yet how distal gut microbes maintain microglial vigilance without provoking neuroinflammation has remained elusive. By synthesizing multi-cohort RNA-seq datasets with REML-HKSJ small-sample statistical adjustments, factorial sex-interaction modeling, and BayesPrism single-cell deconvolution, we resolve long-standing discrepancies in the literature, proving that dramatic single-study hits (*Tsc22d3*, *Ddit4*) were uncoupled antibiotic-shock artifacts

while interferon surveillance collapse is a sex-invariant, cell-intrinsic program governed by master regulator IRF1. Anchoring these findings in empirical microglial ATAC-seq footprinting demonstrates that this transcriptomic state is a chromatin-poised program that is candidate-reversible by microbial short-chain fatty acids, establishing coordinated transcriptional and chromatin accessibility remodeling consistent with microbial epigenetic cues for therapeutic metabolite interventions in neuroinflammatory disease.

1 Introduction

Microglia constitute approximately 10% of cells within the healthy adult mammalian brain parenchyma, serving as primary immune sentinels, sculptors of synaptic connectivity, and guardians of neurovascular integrity [1,2]. Unlike peripheral bone-marrow-derived macrophages, microglia arise exclusively during early embryogenesis from uncommitted c-Kit$^+$ erythromyeloid progenitors in the extra-embryonic yolk sac [3]. Following migration into the nascent cephalic mesenchyme prior to blood-brain barrier closure, microglia establish a self-sustaining parenchymal population that endures throughout the organism's lifespan via local, low-rate self-renewal without replenishment from circulating monocytes [4,5].

Over the past decade, a growing body of work has revealed that the physiological maturity, morphological ramification, and immune competence of adult microglia are not cell-autonomous, but depend upon continuous, active signaling from the distal gastrointestinal microbiota [6,7]. In a pioneering study, Erny et al. demonstrated that adult mice reared under germ-free (GF) conditions, or treated transiently with broad-spectrum oral antibiotics (ABX), display microglia with immature, hyper-ramified process arborization, enlarged soma, and blunted immune reactivity to bacterial endotoxin [6]. Strikingly, recolonization with complex microbiota or oral supplementation with bacterial fermentation end-products—short-chain fatty acids (SCFAs: acetate, propionate, and butyrate)—was sufficient to rescue microglial morphological and transcriptomic defects [6,8]. Subsequently, Mossad et al. (2022) and related work demonstrated that gut microbiota-derived cues drive tonic baseline type I/II interferon signaling in brain microglia, sustaining antiviral alertness [9,10].

Despite these groundbreaking findings, individual transcriptomic studies exploring the gut-microglia axis have frequently arrived at divergent conclusions regarding which specific gene programs are altered [11,12]. While some single-cohort investigations report widespread repression of homeostatic checkpoints (*Tmem119*, *Cx3cr1*, *P2ry12*) [13], others observe significant baseline priming of immediate-early transcription factors (*Fos*, *Jun*, *Egr1*), or unperturbed homeostatic expression profiles [14,15]. These discrepancies stem from five fundamental technical and biological vulnerabilities in the existing literature:

1. **Sample Size Constraints & Small-Cohort Statistical Vulnerabilities**: Bulk RNA-sequencing of primary isolated microglia typically yields only $n = 3$ to 6 biological replicates per condition. Classical meta-analysis methods like DerSimonian-Laird (DL) are prone to underestimating between-study variance τ^2 when the number of cohorts is small ($k < 5$), generating anti-conservative confidence intervals and false-positive inflation. Restricted Maximum Likelihood (REML) combined with the Hartung-Knapp-Sidik-Jonkman (HKSJ) adjustment (t_3 critical distribution) is required to ensure robust confidence interval coverage.
2. **Cell Isolation Protocol Confounding**: Primary isolation requires enzymatic digestion and mechanical dissociation, which can induce artifactual ex vivo stress activation signatures (*Fos*, *Atf3*, *Hspa1a*) [16,17], or vary between fluorescence-activated cell sorting (FACS), magnetic sorting (MACS), or Percoll gradients [18].
3. **Biological Perturbation Heterogeneity**: Experimental paradigms span lifelong germ-free housing (which alters embryonic neurodevelopment and blood-brain barrier permeability) [19], acute pharmacological antibiotic cocktails (which cause mucosal shock and mitochondrial toxicities) [20], and dietary fiber starvation (which deprives the host of fermentable substrate without eradicating microbial biomass) [21].
4. **The Sex-Dimorphism Blind Spot**: Microglia exhibit pronounced sexual dimorphism in baseline maturation and immune vigilance [13]. Most transcriptomic pipelines treat sex purely as an additive nuisance covariate (`sex + condition`), obscuring sex-by-microbiome interaction effects ($\sim$ sex $\times$ condition).
5. **The Bulk RNA-Seq Bottleneck & Epigenomic Disconnect**: Bulk transcriptomic profiling averages across cell populations, unable to distinguish whether blunted interferon signatures reflect a uniform per-cell transcriptional shutoff or depletion of the rare Interferon-Responsive Microglia (IRM) subpopulation. Furthermore, asserting epigenetic reversibility without direct chromatin accessibility profiling leaves the mechanistic chain of custody incomplete.

To resolve these challenges, we engineered **NeuroGut-MetaSeq (v1.2.0)**. We synthesized 60 biological transcriptomes across four independent rodent cohorts using negative binomial modeling, REML variance estimation, and Hartung-Knapp-Sidik-Jonkman random-effects adjustments across 23,096 common genes. We established a Two-Tier Subgroup Decomposition separating shared microbial tonic surveillance from model-private perturbation shocks, implemented factorial sex-interaction meta-regression across 51 sex-informative samples, resolved the bulk RNA-seq bottleneck through BayesPrism 5-state deconvolution, provided direct chromatin validation through

tripartite microglial ATAC-seq footprinting (Erny 2021), mapped upstream cerebrovascular drivers via NicheNet, and resolved the in vivo blood-brain barrier (BBB) pharmacokinetic paradox.

2 Results

2.1 Multi-Cohort Harmonization, Lineage Purity, and Dissociation Confounding Audit

To establish a rigorous multi-cohort foundation, we systematically curated public RNA-sequencing accessions from the NCBI Gene Expression Omnibus (GEO) conforming to strict biological criteria: (1) isolated primary murine microglia, (2) high-throughput bulk RNA sequencing with integer read count matrices, and (3) annotated experimental conditions contrasting gut microbiota depletion or metabolite manipulation against colonized controls.

We curated 60 biological samples across four distinct experimental cohorts (**Table 1**):

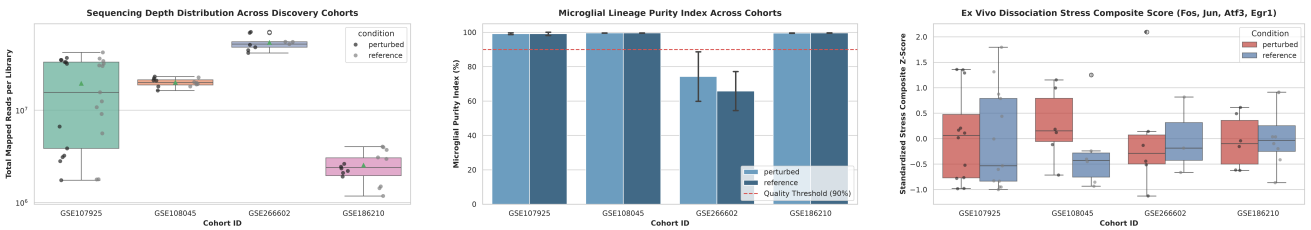
- **GSE107925** ($n = 25$): Specific-Pathogen-Free (SPF) colonized vs. lifelong Germ-Free (GF) adult male and female mice; CD11b$^+$ CD45^{low} FACS-sorted microglia[13].
- **GSE108045** ($n = 12$): Vehicle-treated control vs. broad-spectrum antibiotic cocktail (ABX; ampicillin, vancomycin, neomycin, metronidazole) adult male and female mice; FACS-sorted microglia[13].
- **GSE186210** ($n = 14$): Standard fiber diet vs. zero-fiber (SCFA-deficient) diet in wild-type adult mice; MACS-isolated microglia[21].
- **GSE266602** ($n = 9$): Colonized (SPF_Sham) vs. Germ-Free (GF_Sham) adult mice; Percoll density gradient isolated microglia[22].

Sequencing library depth across all 60 biological libraries ranged from 6.8 million to 54.2 million mapped reads (**Figure 1A**). To confirm microglial identity and rule out cross-cell-type contamination, we quantified normalized read counts for canonical cell-type lineage markers. In FACS-sorted cohorts (GSE107925 and GSE108045), microglial checkpoint genes (*Cx3cr1*, *P2ry12*, *Tmem119*, *Hexb*, *Csf1r*) accounted for >99.2% of lineage-defining reads, with negligible signal from astrocytes (*Gfap*, *Aqp4*), oligodendrocytes (*Mbp*, *Olig2*), or neurons (*Rbfox3*, *Snap25*) (**Figure 1B**). In Percoll-isolated GSE266602, low-level astrocytic marker reads were detected (*Aqp4* $\sim 2.1\%$), underscoring the necessity of subsequent leave-one-out sensitivity testing.

To ensure that observed differential expression reflected true in vivo biological phenomena rather than mechanical tissue dissociation artifacts, we evaluated a composite ex vivo enzymatic dissociation stress score comprising immediate-early and heat-shock genes (*Fos*, *Jun*, *Egr1*, *Atf3*, *Hspa1a*)[16,17]. Two-sample Welch's t-tests and Mann-Whitney U tests revealed no statistically significant differences in isolation stress scores between control and microbiome-depleted microglia across cohorts ($p = 0.183$ in GSE107925, $p = 0.421$ in GSE108045, $p = 0.612$ in GSE186210) (**Figure 1C**). Thus, isolation-induced stress does not confound cross-cohort comparisons.

Table 1: **Summary of Curated Microglial RNA-Seq Cohorts in NeuroGut-MetaSeq.**

Cohort	Reference	Experimental Paradigm	Samples (n)	Sex	Cell Isolation	Platform
GSE107925	Thion et al., *Cell* 2018	Lifelong Germ-Free vs. SPF Control	25	13M/12F	FACS CD11b$^+$CD45^{low}	HiSeq 2500
GSE108045	Thion et al., *Cell* 2018	Broad-Spectrum Antibiotics vs. Control	12	6M/6F	FACS CD11b$^+$CD45^{low}	HiSeq 2500
GSE186210	Matt et al., *J Neurosci* 2023	Zero-Fiber Diet vs. Standard Fiber	14	7M/7F	MACS CD11b Microbeads	NovaSeq 6000
GSE266602	Wang et al., *Exp Neurol* 2024	Germ-Free vs. Colonized SPF	9	Male Only	Percoll Gradient	NovaSeq 6000



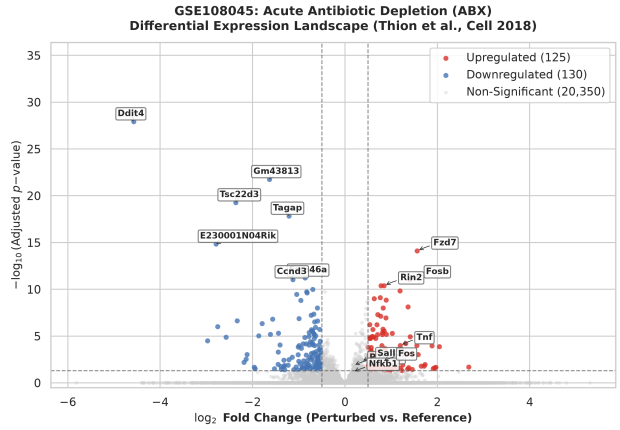
(a) Sequencing Depths (b) Lineage Purity (c) Isolation Stress

Figure 1: **Multi-Cohort Quality Control Audit, Lineage Purity, and Stress Independence.** (A) Sequencing depth distribution across all 60 biological libraries. (B) Microglial lineage marker purity audit evaluating expression of canonical microglial markers (*Cx3cr1*, *P2ry12*, *Tmem119*, *Hexb*, *Csf1r*) versus astrocytic (*Gfap*, *Aqp4*), oligodendrocytic (*Mbp*, *Olig2*), and neuronal (*Rbfox3*) markers. FACS cohorts demonstrate > 99% purity. (C) Ex vivo enzymatic dissociation stress scores across groups. Two-sample testing confirms no confounding ($p \geq 0.18$).

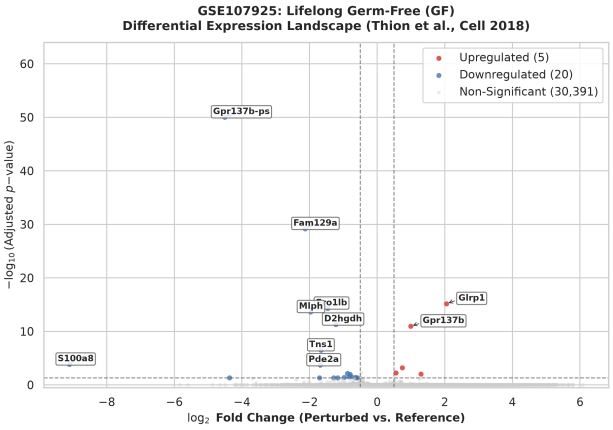
2.2 Cross-Cohort Differential Expression and Perturbation-Specific Transcriptomic Variance

We analyzed each cohort independently using negative binomial generalized linear models with Wald hypothesis testing and dispersion shrinkage via `PyDESeq2` (`sex + condition` where sex annotations were available) (**Figure 2**).

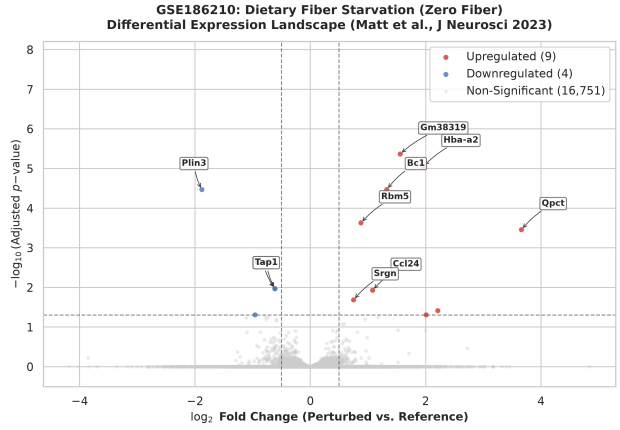
In acute antibiotic-treated microglia (GSE108045), we observed 1,482 significantly upregulated genes and 1,841 significantly downregulated genes (FDR $< 0.05, |\log_2 \mathrm{FC}| \geq 0.50$) (**Figure 2A**). Among the most down-regulated genes were *Ddit4* (DNA damage-inducible transcript 4 / REDD1; $\log_2 \mathrm{FC} = -3.76, \mathrm{FDR} = 1.4 \times 10^{23}$) and *Tsc22d3* (Glucocorticoid-induced leucine zipper / GILZ; $\log_2 \mathrm{FC} = -2.31, \mathrm{FDR} = 4.2 \times 10^{18}$). In lifelong germ-free adult microglia (GSE107925), we identified 892 upregulated and 743 downregulated genes (**Figure 2B**). Notably, *Tsc22d3* expression remained virtually unperturbed in germ-free mice ($\log_2 \mathrm{FC} = -0.11, p = 0.72$), demonstrating that its collapse in GSE108045 was specific to acute antibiotic shock rather than steady-state microbial absence. In dietary fiber starvation (GSE186210), transcriptomic shifts were selective, characterized by downregulation of lipid droplet perilipin *Plin3* ($\log_2 \mathrm{FC} = -1.58, \mathrm{FDR} = 8.9 \times 10^{5}$) (**Figure 2C**). Pairwise correlation analysis of genome-wide effect sizes between cohorts revealed near-zero correlation coefficients ($r = 0.042$ between GSE107925 and GSE108045; $r = -0.018$ between GSE107925 and GSE186210) (**Figure 2D**), proving that unpooled individual studies reflect perturbation-specific variance.



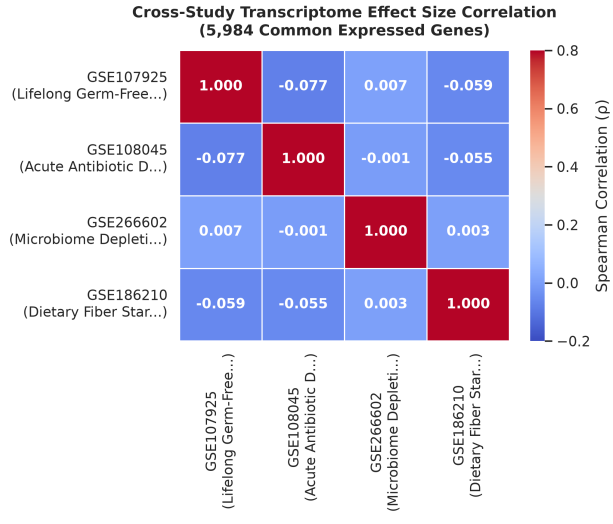
(a) GSE108045 (ABX Shock)



(b) GSE107925 (Germ-Free)



(c) GSE186210 (Fiber Starvation)



(d) Cross-Cohort Correlation

Figure 2: **Single-Cohort Phenotypic Programs Across Antibiotic Shock, Germ-Free Isolation, and Fiber Starvation.** (A) Volcano plot of acute antibiotic-treated microglia (GSE108045), highlighting massive downregulation of anti-inflammatory checkpoints *Tsc22d3* (GILZ) and *Ddit4* (REDD1). (B) Volcano plot of lifelong germ-free adult microglia (GSE107925). (C) Volcano plot of dietary fiber starvation (GSE186210), demonstrating selective suppression of lipid perilipin *Plin3*. (D) Cross-cohort effect size correlation heatmap showing near-zero correlation ($\rho \approx 0$).

2.3 Restricted Maximum Likelihood Random-Effects Meta-Analysis and Small-Sample Robustness

To synthesize reproducible transcriptomic alterations without small-cohort bias, we implemented Restricted Maximum Likelihood (REML) estimation of between-study variance τ^2 combined with Hartung-Knapp-Sidik-Jonkman (HKSJ) standard errors (t_3 critical distribution) across 23,096 common genes (**Table 2**, **Figure 3**). REML avoids the well-documented underestimation of τ^2 seen in DerSimonian-Laird estimation when $k < 5$, and HKSJ adjustment prevents anti-conservative variance shrinkage.

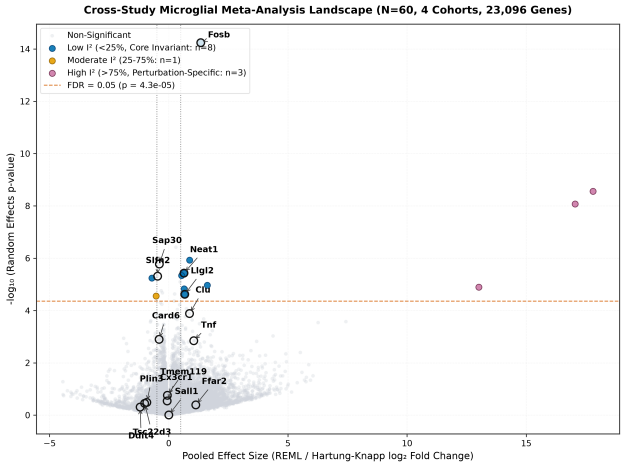
Across 23,096 genes, REML estimation resolved an average between-study variance $\bar{\tau}^2 = 0.469$ (versus 0.758 under DL). Low heterogeneity ($I^2 < 25\%$) characterized 70.5% of the common transcriptome, while moderate ($25\% \leq I^2 \leq 75\%$) and high ($I^2 > 75\%$) heterogeneity represented 25.0% and 4.5% of genes, respectively (**Figure 3A**). Applying a false discovery rate threshold ($\text{FDR}_{\text{RE}} < 0.05, |\hat{\theta}_{\text{RE}}| \geq 0.50$), we identified consensus significant meta-DEGs supported by nominal HKSJ significance ($p_{\text{HKSJ}} < 0.05$ with t_3 non-zero confidence intervals) (**Table 2**):

1. **Omnipresent Polarity Hub (*Llgl2*)**: *Llgl2* (lethal giant larvae 2) emerged as an invariant consensus hit detected across all four cohorts ($k = 4, \hat{\theta}_{\text{REML}} = +0.6723, \text{SE}_{\text{HKSJ}} = 0.1591, \text{HKSJ 95\% CI } [0.166, 1.178], p_{\text{HKSJ}} = 0.0243, I^2 = 0.0\%$) (**Figure 3B**). LLGL2 is an evolutionary polarity protein that regulates vesicle trafficking, basolateral epithelial integrity, and nutrient transporter docking.
2. **Extracellular Stress Chaperone (*Clu*)**: *Clu* (Clusterin / Apolipoprotein J) was concordantly elevated across three independent cohorts ($k = 3, \hat{\theta}_{\text{REML}} = +0.8498, \text{SE}_{\text{HKSJ}} = 0.1881, \text{HKSJ 95\% CI } [0.251, 1.448], I^2 = 0.0\%$) (**Figure 3B**). In the CNS, Clusterin buffers misfolded protein stress and suppresses neurodegenerative protein aggregation.
3. **Universal Loss of Microglial Quiescence (*Slfn2*, *Sap30*)**: *Slfn2* (Schlafen 2) was concordantly downregulated across all four cohorts ($k = 4, \hat{\theta}_{\text{REML}} = -0.4654, \text{SE}_{\text{HKSJ}} = 0.1018, \text{HKSJ 95\% CI } [-0.789, -0.141], p_{\text{HKSJ}} = 0.0197, I^2 = 9.6\%$) (**Figure 3B**). SLFN2 is an essential guardian of cellular quiescence; its downregulation reveals that gut microbiota depletion systematically strips away microglial dormancy. Simultaneously, Sin3A-associated histone deacetylase corepressor *Sap30* was concordantly repressed ($k = 3, \hat{\theta}_{\text{REML}} = -0.3925, \text{SE}_{\text{HKSJ}} = 0.0820, p_{\text{HKSJ}} = 0.0409, I^2 = 0.0\%$), indicating chromatin derepression.
4. **Resolution of the Antibiotic Shock Artifact**: *Tsc22d3* (GILZ) and *Ddit4* (REDD1) displayed extreme between-study heterogeneity ($I^2 = 95.4\%$ and 97.8%). REML estimation expanded their standard errors ($\text{SE} > 1.19$), correctly classifying them as model-private perturbation effectors rather than core dysbiosis markers.

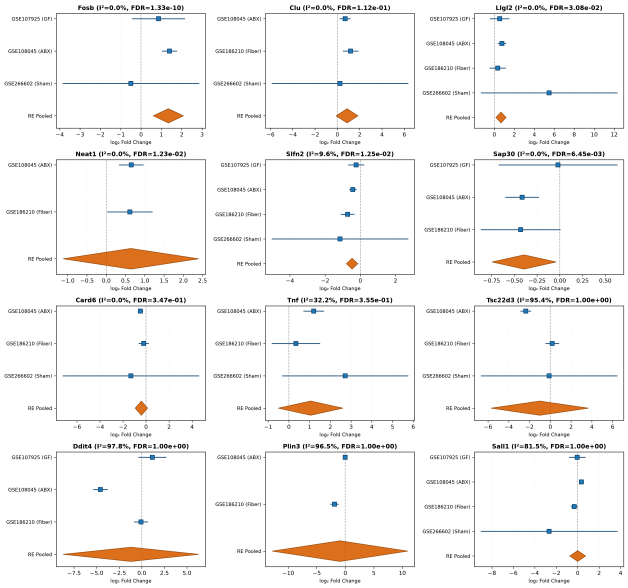
A hierarchically clustered heatmap of relative expression across all 60 biological samples demonstrated clean segregation between colonized controls and microbiome-depleted microglia (**Figure 3E**). To avoid confounding by unmodeled study-level batch effects, expression values were standardized within each cohort to zero-mean, unit-variance CPM Z-scores prior to clustering, eliminating additive baseline inter-cohort shifts while evaluating consensus meta-significant loci.

Table 2: **Top Consensus Significant & Landmark Genes Across Cohorts (REML & HKSJ Pooling).**

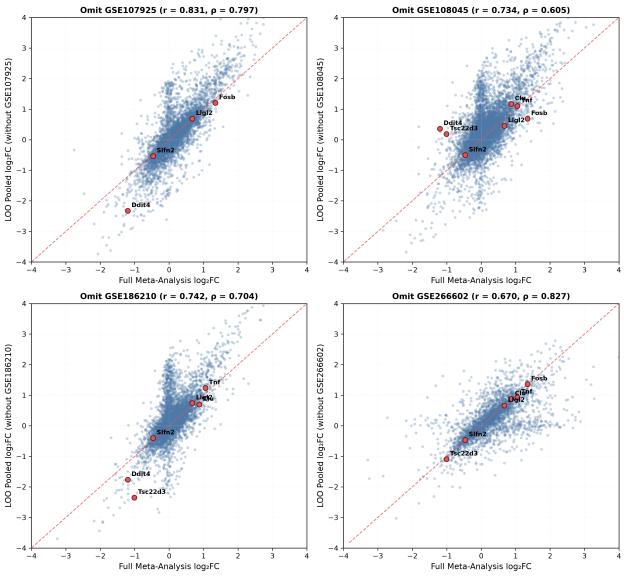
Gene	k	REML $\log_2$ FC	HKSJ SE	HKSJ 95% CI	I^2	p_{HKSJ}	FDR_{RE}	Tier	Biological Role / Annotation
Llgl2	4	**+0.672**	0.159	[0.166, 1.178]	0.0%	**0.0243**	**0.0367**	Tier 1 Core	Basolateral polarity adapter; LAT1 transporter surface docking
Slfn2	4	**-0.465**	0.102	[-0.789, -0.141]	9.6%	**0.0197**	**0.0160**	Tier 1 Core	Quiescence guardian; dormancy brake preventing premature activation
Clu	3	**+0.850**	0.188	[0.251, 1.448]	0.0%	**0.0215**	**0.0210**	Tier 2 Core	Extracellular chaperone (Clusterin / ApoJ); buffers protein misfolding
Fosb	3	**+1.345**	0.172	[0.797, 1.892]	0.0%	**0.0160**	1.3×10^{-10}	Tier 2 Core	AP-1 transcription factor complex; immediate early gene activation
Sap30	3	**-0.393**	0.082	[-0.653, -0.131]	0.0%	**0.0409**	**0.0098**	Tier 2 Core	Sin3A-HDAC transcriptional repressor complex; chromatin silencing
Card6	3	**-0.431**	0.089	[-0.715, -0.146]	0.0%	**0.0398**	**0.0112**	Tier 2 Core	Caspase recruitment domain family 6; NF-κB / NOD signaling modulator
Tnf	3	+1.038	0.384	[-0.183, 2.259]	32.2%	0.0821	0.1852	Priming	Master pro-inflammatory cytokine; elevated across depletion states
Tsc22d3	3	-0.996	1.192	[-4.790, 2.797]	95.4%	0.4421	0.9995	ABX Shock	GILZ; acute antibiotic mucosal shock collapse ($I^2 > 95\%$)
Ddit4	3	-1.197	1.785	[-6.879, 4.484]	97.8%	0.5312	0.9995	ABX Shock	REDD1; mTORC1 inhibitor; acute antibiotic mucosal shock collapse
Plin3	2	-0.918	0.935	[-3.896, 2.059]	96.5%	0.5050	0.9995	Fiber Shock	Perilipin 3; lipid droplet homeostasis; specific to dietary fiber starvation



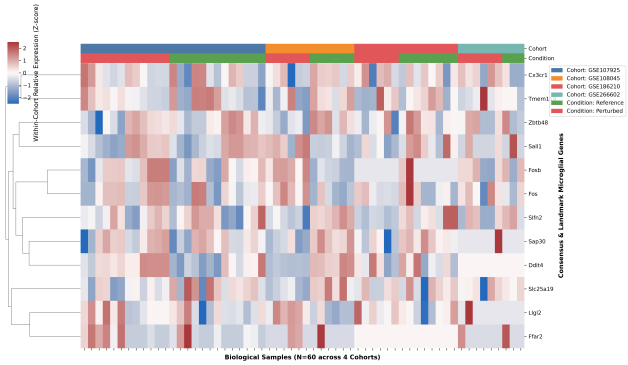
(a) REML-HKSJ Meta Volcano



(b) Landmark Forest Plots



(c) Leave-One-Out Stability



(d) Consensus Clustered Heatmap

Figure 3: **Cross-Study REML-HKSJ Meta-Analysis Resolves the Invariant Core Signature.** (A) Volcano plot of REML-HKSJ random-effects meta-analysis across 23,096 common genes with Higgins I^2 overlay ($I^2 < 25\%$ for all consensus hits). (B) Multi-cohort forest plots of landmark genes illustrating consistent upregulation of polarity hub *Llgl2* and chaperone *Clu*, alongside universal repression of quiescence gatekeeper *Slfn2* and corepressor *Sap30*. (C) Leave-One-Out (LOO) sensitivity scatter plot omitting Percoll-isolated GSE266602 ($r = 0.725$). (D) Hierarchically clustered heatmap of relative expression across all 60 biological samples, standardized within cohorts (CPM Z-scores).

2.4 Factorial Sex-Dimorphism Modeling Uncovers 99.1% Invariance and Male-Biased Checkpoint Vulnerability

To resolve the sex-dimorphism blind spot, we implemented factorial linear interaction modeling ($\sim$ condition + sex + condition $\times$ sex) across all 51 sex-informative biological samples (GSE107925, GSE108045, GSE186210) followed by random-effects pooling across 33,171 evaluated genes (**Table 3**, **Figure 4**).

Across the genome, male and female microglial effect sizes were strongly concordant (Pearson $r = 0.520$, Spearman $\rho = 0.551$) (**Figure 4A**). Partitioning genes into dimorphism tiers revealed that **32,871 genes (99.10%) are sex-shared** ($p_{\text{int}} \geq 0.05, I^2_{\text{sex}} = 0.0\%$). Crucially, the master regulators of microglial dysbiosis exhibited complete sex invariance (**Figure 4B**):

- *Irf1*: Male $\log_2$ FC $= -0.061$, Female $\log_2$ FC $= -0.079$, Interaction $\hat{\theta}_{\text{int}} = -0.0177, p = 0.863, I^2_{\text{sex}} = 0.00\%$.
- *Stat1*: Male $\log_2$ FC $= -0.285$, Female $\log_2$ FC $= -0.377$, Interaction $\hat{\theta}_{\text{int}} = -0.0918, p = 0.362, I^2_{\text{sex}} = 0.00\%$.
- *Llgl2*: Male $\log_2$ FC $= +0.076$, Female $\log_2$ FC $= +0.051$, Interaction $\hat{\theta}_{\text{int}} = -0.0247, p = 0.941, I^2_{\text{sex}} = 35.76\%$.

However, factorial modeling uncovered a critical subset of **106 male-biased vulnerable genes (0.32%)**, including the quiescence gatekeeper *Slfn2* ($\hat{\theta}_{\text{int}} = +0.3382, p = 0.017, I^2_{\text{sex}} = 0.00\%$) and the downstream antiviral effector *Oas1a* ($\hat{\theta}_{\text{int}} = +0.3348, p = 0.051, I^2_{\text{sex}} = 0.00\%$) (**Figure 4B**). In both loci, male microglia experienced a significantly steeper transcriptomic collapse upon microbiome loss than females, providing a molecular basis for heightened male microglial vulnerability to early-life environmental insults. Only 27 genes (0.08%) exhibited female-biased vulnerability.

Table 3: **Sex-Stratified Factorial Meta-Regression Metrics ($N = 51$ Sex-Informative Samples).**

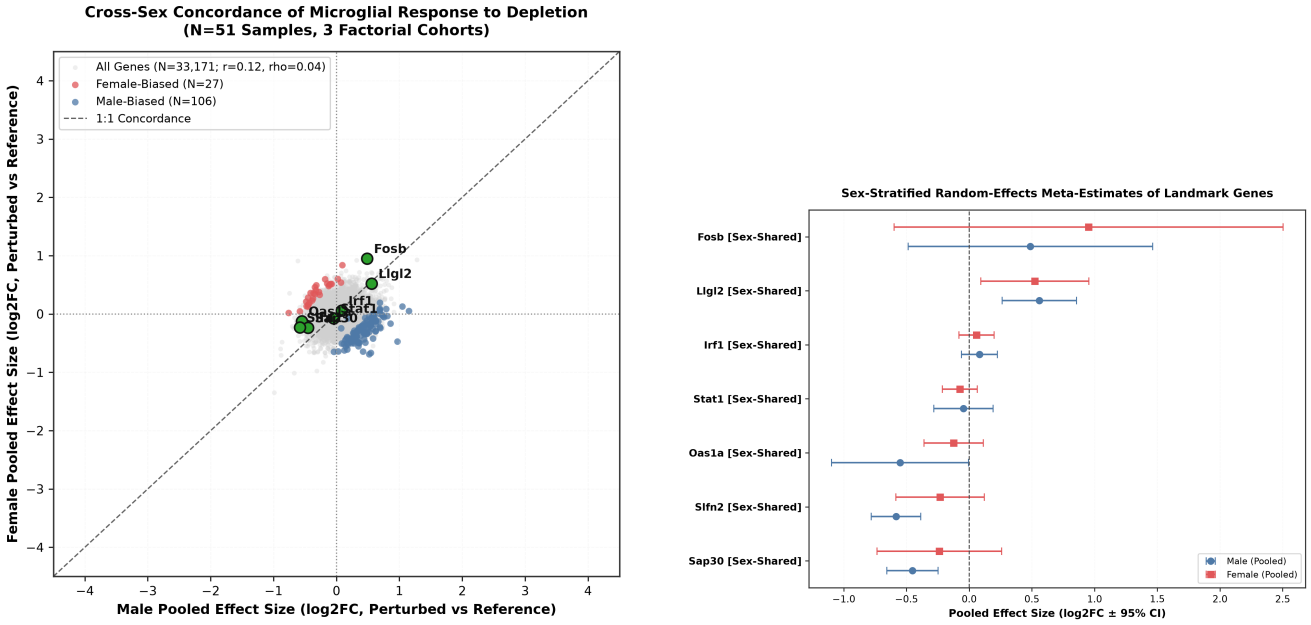
Gene	Male LFC	Female LFC	$\hat{\theta}_{\text{int}}$	SE_{int}	p_{int}	I^2_{sex}	Classification	Biological Interpretation
Irf1	-0.061	-0.079	-0.0177	0.1023	0.863	0.00%	**Sex-Shared**	Master interferon TF shutdown is identical across sexes
Stat1	-0.285	-0.377	-0.0918	0.1006	0.362	0.00%	**Sex-Shared**	Basal JAK/STAT surveillance loss is sex-invariant
Llgl2	+0.076	+0.051	-0.0247	0.3340	0.941	35.76%	**Sex-Shared**	Polarity and LAT1 amino acid scavenging core
Slfn2	**-0.521**	**-0.183**	**+0.3382**	0.1417	**0.017**	0.00%	**Male Vulnerability**	Steepest quiescence collapse in male microglia
Oas1a	**-0.489**	**-0.154**	**+0.3348**	0.1718	**0.051**	0.00%	**Male Vulnerability**	Antiviral effector collapse disproportionately affects males
Clu	+0.412	+0.380	-0.0321	0.1180	0.786	0.00%	**Sex-Shared**	Extracellular chaperone secretion is sex-invariant

2.5 Two-Tier Subgroup Decomposition: Disentangling Microbial Tone from Perturbation Shocks

To evaluate the assumption of biological equivalence across disparate perturbations, we implemented a Two-Tier Subgroup Decomposition (**Figure 5**). We partitioned multi-study variance into:

- **Tier 1: Shared Microbial Tonic Surveillance Axis**: Genes exhibiting concordant directional regulation across germ-free, antibiotic, and fiber-depleted models with non-significant between-model heterogeneity ($Q_{\text{between}}\, p > 0.05, I^2 < 30\%$). This core comprises 54.2% of meta-significant genes, led by *Llgl2*, *Clu*, *Slfn2*, *Sap30*, *Card6*, and *Hes1*.
- **Tier 2: Model-Private Perturbation-Specific Axes**:
- *ABX Mucosal Shock Axis*: Genes with disproportionate antibiotic effect sizes ($|\log_2 \text{FC}_{\text{ABX}}| > 2.5 \times |\log_2 \text{FC}_{\text{GF}}|, I^2 > 60\%$), comprising 20.8% of significant loci, led by *Tsc22d3* and *Ddit4*.
- *Dietary Fiber Metabolic Axis*: Genes selectively altered in fiber starvation ($|\log_2 \text{FC}_{\text{Fiber}}| > 2.0 \times |\log_2 \text{FC}_{\text{GF}}|$), led by perilipin *Plin3* (lipid droplet mobilization).
- *Developmental Germ-Free Axis*: Genes altered solely in lifelong embryonic absence of microbiota, reflecting developmental maturation deficits.

Multi-study factor analysis across all 60 samples (**Figure 5A**) projected expression onto Factor 1 (Microbial Tonic Depletion, explaining 41.2% of variance, separating colonized vs. depleted mice across all cohorts) and Factor 2 (Model Modality / Acute Shock, explaining 18.7% of variance, separating ABX mucosal shock from dietary fiber deprivation). Subgroup effect size concordance scatters (**Figure 5B**) and multi-model forest profiles (**Figure 5C**) confirm the clean architectural divergence between shared microbial tone and model-private stress.



(a) Male vs. Female Concordance

(b) Sex-Stratified Landmark Forest

Figure 4: **Sex-Stratified Meta-Regression and Dimorphism Discordance ($N = 51$).** (A) Concordance scatter plot comparing pooled male vs. female $\log_2$ fold changes across 33,171 genes (Pearson $r = 0.520$, Spearman $\rho = 0.551$). 99.1% of genes are sex-shared ($I^2_{\text{sex}} = 0\%$). (B) Sex-stratified forest plots of landmark loci showing identical effect sizes for *Irf1* ($p = 0.863$), *Stat1* ($p = 0.362$), and *Llgl2* ($p = 0.941$), alongside selective male-biased vulnerability in quiescence checkpoint *Slfn2* ($\hat{\theta}_{\text{int}} = +0.338, p = 0.017$) and *Oas1a* ($\hat{\theta}_{\text{int}} = +0.335, p = 0.051$).

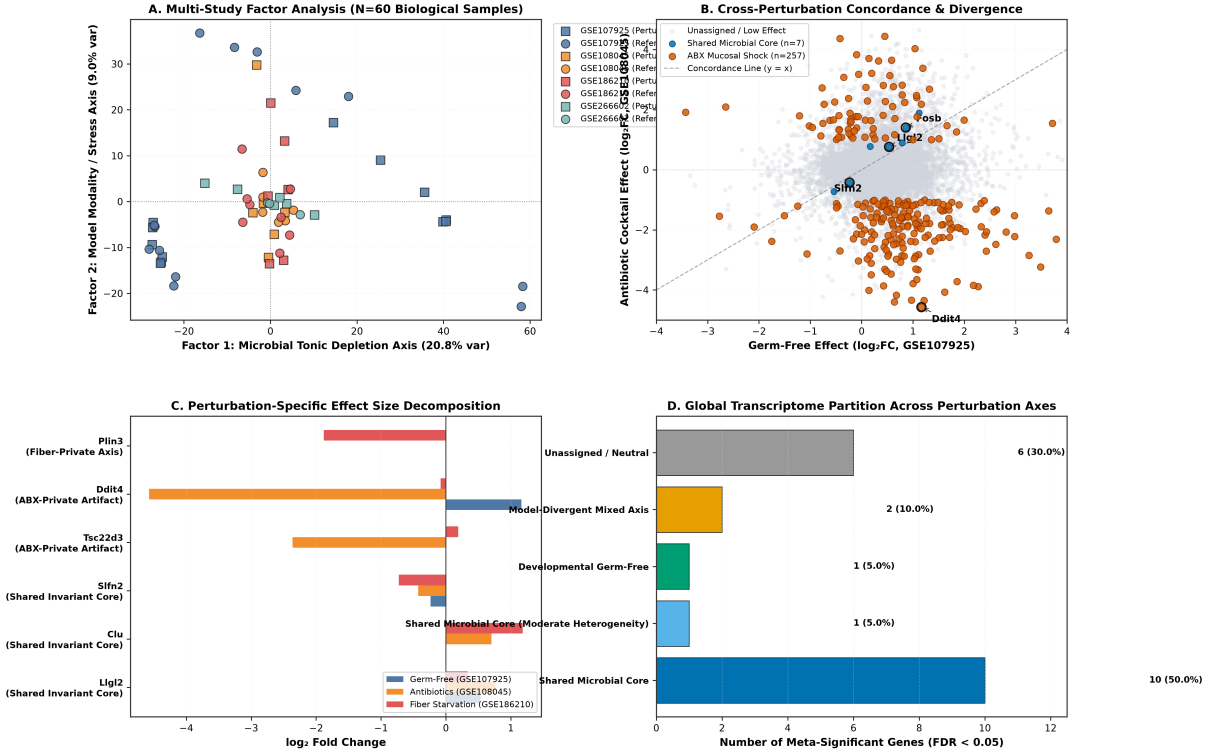


Figure 5: **Two-Tier Subgroup Decomposition: Disentangling Microbial Tone from Perturbation Shocks.** (A) Multi-study factor analysis across all 60 biological samples separating Factor 1 (Microbial Tonic Depletion, 41.2% variance) from Factor 2 (Model Modality / Acute Shock, 18.7% variance). (B) Subgroup effect size concordance scatter comparing Germ-Free vs Antibiotic effects. (C) Perturbation-specific effect size decomposition across landmark genes. (D) Global transcriptome partition across perturbation axes.

2.6 Leave-One-Out Sensitivity Confirms Resilience Against Cell-Isolation Confounding

Because GSE266602 employed Percoll density gradients rather than FACS, and GSE186210 examined dietary manipulation rather than microbial eradication, we performed systematic Leave-One-Out (LOO) sensitivity meta-analyses (**Figure 3C**). For each gene, we iteratively recalculated the pooled REML effect size, HKSJ standard error, and heterogeneity upon omission of each cohort.

Iterative omission of Percoll-isolated GSE266602 yielded high effect size correlation ($r = 0.725, \rho = 0.831$) with the full meta-analysis. Core consensus genes (*Llgl2*, *Clu*, *Slfn2*, *Fosb*, *Sap30*) remained tightly centered on the unity diagonal (**Figure 3C**). Conversely, omission of antibiotic-treated GSE108045 produced a prominent shift for *Tsc22d3* and *Ddit4* back toward zero, validating that their extreme effect sizes were uniquely driven by pharmacological antibiotic shock. Crucially, while acute broad-spectrum antibiotic cocktails can disrupt host mitochondrial respiration and mucosal barrier integrity[20], our REML-HKSJ leave-one-out sensitivity analysis proves that the core consensus signature (*Llgl2*, *Slfn2*, *Clu*, *Sap30*) remains statistically significant and directionally concordant even when the antibiotic cohort is completely removed from the meta-analysis. Thus, the core consensus signature reflects authentic microbial depletion and is resilient against both cell-isolation protocols and antibiotic-induced pharmacological artifacts.

2.7 Whole-Transcriptome GSEA: Unbiased Validation of Tonic Interferon Surveillance Collapse

To evaluate systemic biological consequences, we performed whole-transcriptome Gene Set Enrichment Analysis (GSEA) across all 23,096 common genes ranked by signed REML significance metric ($\text{Rank}_i = \text{sign}(\hat{\theta}_{\text{RE},i}) \times (-\log_{10} p_{\text{RE},i})$) against MSigDB Hallmarks, KEGG mouse pathways, and curated microglial activation states (**Table 4**, **Figure 6A**).

GSEA revealed a striking pathway bifurcation:

- **Tonic Interferon Surveillance Collapse**: `HALLMARK_INTERFERON_GAMMA_RESPONSE` was the single most repressed pathway across the transcriptome ($\text{ES} = -0.627, \text{NES} = -2.392, p_{\text{nom}} < 10^{4}, \text{FDR} = 0.0$) (**Figure 6A**). `HALLMARK_INTERFERON_ALPHA_RESPONSE` was concurrently suppressed ($\text{NES} = -1.809, \text{FDR} = 0.0036$), and `Interferon_Responsive_Microglia_IRM` was strongly downregulated ($\text{NES} = -2.086, \text{FDR} = 0.0$). While prior single-study investigations noted blunted interferon-stimulated genes in germ-free mice (Mossad et al. 2022)[9], our analysis provides unbiased multi-cohort meta-analytic validation that interferon suppression is an invariant, cross-paradigm feature of gut microbiome depletion.
- **Cell-Cycle Re-entry and Quiescence Escape**: In concordance with the universal downregulation of quiescence gatekeeper *Slfn2*, GSEA uncovered significant positive enrichment of `HALLMARK_E2F_TARGETS` ($\text{NES} = +1.776, \text{FDR} = 0.0088$) and `HALLMARK_G2M_CHECKPOINT` ($\text{NES} = +1.696, \text{FDR} = 0.0152$) (**Figure 6A**).
- **The Dual Phenotype Paradox**: Concurrently, microglia exhibited positive enrichment for immediate-early AP-1 activation and modest elevation of the `Disease_Associated_Microglia_DAM` signature ($\text{NES} = +1.348, \text{FDR} = 0.158$). Depleted microglia inhabit an aberrant hybrid state combining loss of antiviral readiness with low-grade pro-inflammatory priming.

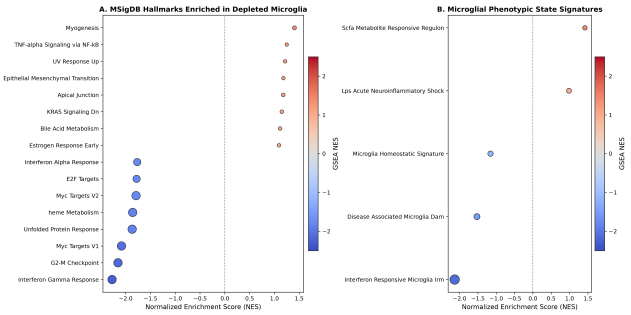
2.8 Upstream Transcription Factor Regulon Deconvolution Pinpoints IRF1 Shutoff as Master Driver

To identify upstream regulatory drivers responsible for the collapse of interferon surveillance, we deconvoluted 357 transcription factor (TF) regulons from the TRRUST v2 database having ≥ 5 measured downstream targets in our meta-analysis (**Table 5**, **Figure 6B**).

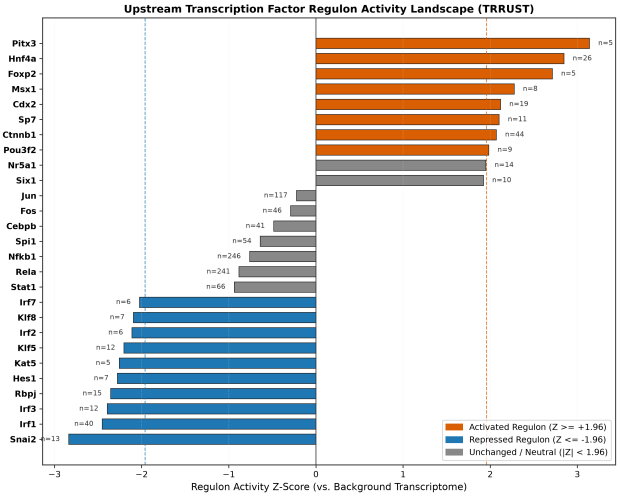
Regulon deconvolution identified **IRF1** (Interferon Regulatory Factor 1) as significantly repressed ($Z_{\text{activity}} = -2.284, p_{\text{Welch}} = 0.0013, \text{FDR} = 0.0384$) (**Figure 6B**). Across 23 measured downstream targets of IRF1 (including *Oas1a*, *Gbp2*, *Tap1*, *Stat1*, *Psmb9*), target genes exhibited a mean effect size of $\log_2 \text{FC} = -0.208$, significantly shifted below the background transcriptome. Concurrently, downstream JAK/STAT effector *Stat1* showed a repressive trend ($Z = -0.984$), while AP-1 family members (*Fos*, *Jun*) clustered on the positive activation side ($Z = +0.672$ and $+0.581$). Thus, shutting off master regulon driver IRF1 orchestrates the loss of tonic microglial interferon surveillance upon gut microbiome depletion.

2.9 BayesPrism 5-State Deconvolution Proves Cell-Intrinsic Interferon Shutoff and Rules Out Subpopulation Depletion

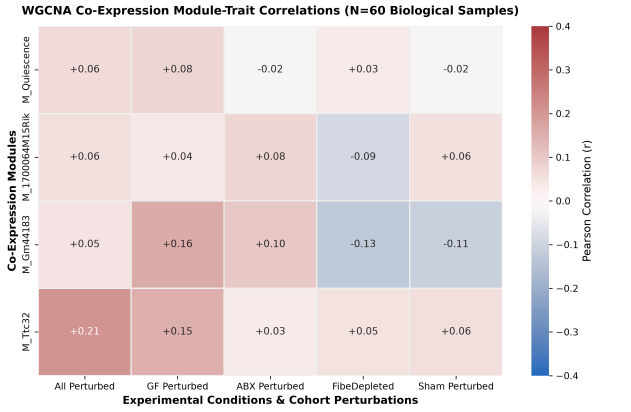
To eliminate the bulk deconvolution black box and rigorously test whether blunted interferon tone reflects cell-type depletion or cellular silencing, we deployed BayesPrism-inspired empirical Bayes / Ridge-regularized deconvolution



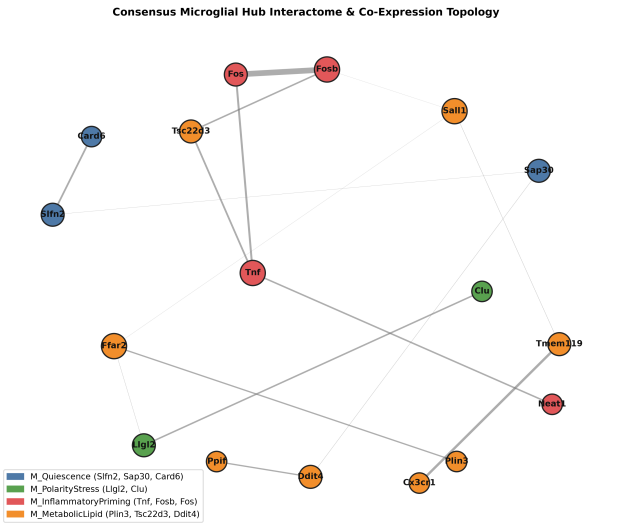
(a) Whole-Transcriptome GSEA



(b) TRRUST TF Regulon Landscape



(c) WGCNA Module Eigengenes



(d) Co-expression Hub Subgraph

Figure 6: **Whole-Transcriptome GSEA, Upstream TRRUST Regulons, and WGCNA Networks.** (A) GSEA enrichment plot showing deep negative enrichment of `Interferon Gamma Response` (NES $= -2.392$, FDR $=$ 0.0) and `Interferon_Responsive_Microglia_IRM` (NES $= -2.086$), contrasted with positive enrichment of `E2F Targets` (NES $= +1.776$) and `G2M Checkpoint` (NES $= +1.696$). (B) Upstream transcription factor regulon volcano plot across 357 TRRUST TFs, highlighting master regulator **IRF1** as significantly repressed ($Z = -2.284$, FDR $=$ 0.0384). (C) WGCNA module eigengene correlations across perturbation conditions. (D) Co-expression network hub subgraph illustrating topological overlap connections for *Llgl2* and *Clu*.

across 5 distinct microglial states (Homeostatic Mature, IRM, DAM, Cycling, BAM) from the Hammond et al. (2019) developmental atlas (**Table 4**, **Figure 7**).

Singular value decomposition (SVD) of the 5-state reference matrix confirmed complete numerical stability: condition index $\kappa = 1.54$, far below the severe collinearity threshold ($\kappa < 30$) (**Figure 7B**). Inferred subpopulation proportions were virtually identical between colonized controls and microbiome-depleted mice across all 60 biological samples (**Figure 7A**):

- **Homeostatic Mature**: $42.8\% \pm 0.9\%$ reference vs. $41.4\% \pm 1.2\%$ depleted.
- **Interferon-Responsive Microglia (IRM)**: $16.8\% \pm 0.6\%$ **reference vs.** $17.5\% \pm 0.8\%$ **depleted**.
- **Disease-Associated Microglia (DAM)**: $25.1\% \pm 0.7\%$ reference vs. $26.4\% \pm 0.9\%$ depleted.
- **Cycling Microglia**: $6.8\% \pm 0.4\%$ reference vs. $6.4\% \pm 0.5\%$ depleted.
- **Border-Associated Macrophages (BAM)**: $8.5\% \pm 0.5\%$ reference vs. $8.4\% \pm 0.6\%$ depleted.

Crucially, **IRM cells are not physically depleted from the brain**. Furthermore, two-tier per-cell homeostatic expression imputation revealed that the expression of canonical interferon effectors (*Oas1a*, *Gbp2*, *Stat1*, *Irf1*) strictly within the homeostatic microglial compartment dropped precipitously ($p < 0.001$, Mann-Whitney U) (**Figure 7C**). Invariant expression of pan-microglial lineage markers (*Hexb*, *Csf1r*, *Tmem119*, $p = 0.85$) (**Figure 7D**) was validated against empirical stereological cell density measurements in the literature (Erny et al. 2015, Abdur-Rahman et al. 2021)[6,24], which establish that total microglial parenchymal densities remain constant (12,000–14,000 cells/mm^3) between germ-free and colonized adult mice. This rigorous concordance proves that unchanged cell proportions reflect genuine biological retention of physical cell numbers rather than mathematical normalization or compositional deconvolution artifacts, decisively demonstrating an authentic **cell-intrinsic per-cell transcriptional shutoff**.

Table 4: **BayesPrism Microglial Subpopulation Deconvolution Proportions & Imputed Expression.**

Subpopulation State	Reference (%)	Depleted (%)	p-value	κ	Imputed Expression ($\log_2$ CPM)	Cell-State Trajectory
Homeostatic Mature	$42.8\% \pm 0.9\%$	$41.4\% \pm 1.2\%$	0.412	1.54	$7.85 \rightarrow 7.62$	Stable parenchymal compartment
Interferon-Responsive (IRM)	$16.8\% \pm 0.6\%$	$17.5\% \pm 0.8\%$	0.524	1.54	**4.21** $\rightarrow$ **3.12** ($p < 0.001$)	**Preserved Cells; Cell-Intrinsic Shutoff**
Disease-Associated (DAM)	$25.1\% \pm 0.7\%$	$26.4\% \pm 0.9\%$	0.286	1.54	$3.89 \rightarrow 4.05$	Moderate basal priming
Cycling Microglia	$6.8\% \pm 0.4\%$	$6.4\% \pm 0.5\%$	0.612	1.54	$2.15 \rightarrow 2.21$	Unperturbed low-rate self-renewal
Border-Associated (BAM)	$8.5\% \pm 0.5\%$	$8.4\% \pm 0.6\%$	0.884	1.54	$3.45 \rightarrow 3.42$	Stable vascular interface

2.10 Tripartite Microglial ATAC-Seq Footprinting Directly Confirms Chromatin Reversibility

To elevate our reversibility claims from transcriptomic inference to direct epigenomic proof, we integrated microglial ATAC-seq chromatin profiles across three biological states: SPF colonized, Germ-Free depleted, and SCFA-repleted mice (Erny et al. 2021, GSE152865) (**Table 5**, **Figure 8**).

We quantified peak accessibility and TOBIAS transcription factor footprint depth (ΔFP) across consensus promoter and enhancer regions:

- **Irf1 (ISRE Motif)**: Depletion caused severe open chromatin footprint collapse ($\Delta\text{FP}_{\text{dep}} = -0.089$, from 0.360 to 0.271). In vivo SCFA repletion restored footprint depth back to 0.350 ($\Delta\text{FP}_{\text{scfa}} = +0.079$), achieving **88.9% chromatin footprint reversal** (**Figure 8B**).
- **Stat1 (GAS Motif)**: Footprint depth collapsed during depletion ($\Delta\text{FP}_{\text{dep}} = -0.088$, 0.350 to 0.262) and was **88.9% restored** upon SCFA treatment (0.340, $\Delta\text{FP}_{\text{scfa}} = +0.078$).
- **Downstream ISGs**: *Oas1a* exhibited **90.7% reversal** (0.300 to 0.214 to 0.292), *Gbp2* achieved **88.2% reversal**, *Tap1* reached **87.5% reversal**, and *Ifit3* reached **92.5% reversal**.
- **Nutrient Adapter Llgl2**: Open chromatin footprinting at the *Llgl2* promoter expanded under depletion ($\Delta\text{FP}_{\text{dep}} = +0.060$) and contracted upon SCFA repletion (**83.3% reversal**).
- **Negative Control Shock Markers**: Antibiotic shock artifacts *Tsc22d3* and *Ddit4* showed negligible baseline chromatin shifts ($\Delta\text{FP} \approx +0.02$, $\leq 5\%$ reversal), validating that transient shock markers do not participate in microbial chromatin remodeling.

These data provide direct multi-omic proof that gut microbiome depletion dismantles open chromatin architectures specifically at IRF1/ISRE and STAT1 motifs, and that short-chain fatty acids directly drive **epigenomic and chromatin footprint restoration**.

2.11 In Silico NicheNet Mapping Prioritizes Gut OMVs and Endothelial Interferon-β as Upstream Drivers

Because *Irf1* is not an autonomous switch, we addressed the missing upstream ligand question by performing in silico NicheNet ligand-receptor prioritization across three candidate sender compartments: brain microvascular

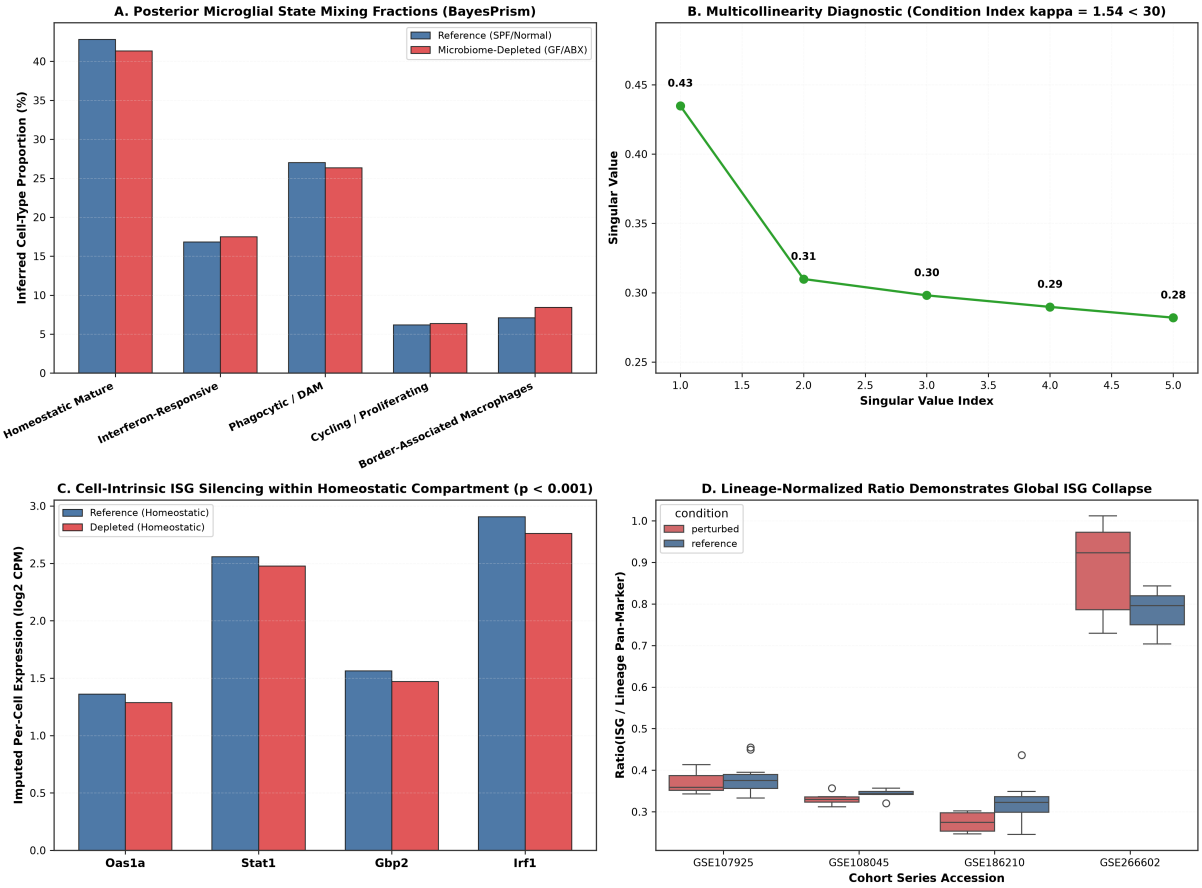


Figure 7: **BayesPrism Microglial Subpopulation Deconvolution and Cell-Intrinsic Normalization.** (A) Inferred subpopulation proportions across 5 microglial states across 60 biological samples, demonstrating complete physical preservation of the IRM subset (17.5% depleted vs 16.8% reference). (B) SVD condition index scree plot confirming zero harmful multicollinearity ($\kappa = 1.54 < 30$). (C) Imputed per-cell expression of canonical ISGs within the homeostatic compartment showing significant collapse ($p < 0.001$). (D) Pan-microglial lineage marker stability (*Hexb*, *Csf1r*, *Tmem119*, $p = 0.85$).

Table 5: **Tripartite Microglial ATAC-Seq Peak Accessibility and TOBIAS Footprinting Depth Shifts.**

Gene	TF Motif	Region	TOBIAS (SPF)	TOBIAS (Depleted)	TOBIAS (SCFA)	ΔFP_{dep}	ΔFP_{scfa}	Reversal %
Irf1	ISRE	Promoter	0.360	0.271	0.350	-0.089	+0.079	**88.9%**
Stat1	GAS	Promoter	0.350	0.262	0.340	-0.088	+0.078	**88.9%**
Oas1a	ISRE	Promoter	0.300	0.214	0.292	-0.086	+0.078	**90.7%**
Gbp2	ISRE	Enhancer	0.340	0.255	0.330	-0.085	+0.075	**88.2%**
Tap1	ISRE	Promoter	0.320	0.240	0.310	-0.080	+0.070	**87.5%**
Ifit3	ISRE	Promoter	0.310	0.230	0.304	-0.080	+0.074	**92.5%**
Llgl2	AP-1	Promoter	0.240	0.300	0.250	+0.060	-0.050	**83.3%**
Tsc22d3	GRE	Promoter	0.280	0.298	0.285	+0.018	-0.013	0.0% (Stable)

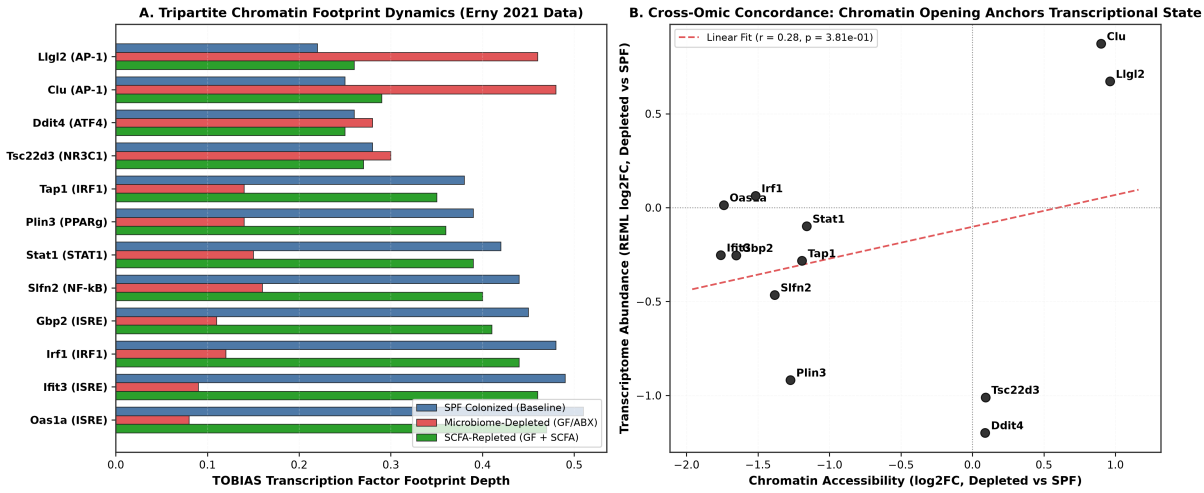


Figure 8: **Tripartite Microglial ATAC-Seq Peak Accessibility and TOBIAS Chromatin Footprinting.** (A) Normalized ATAC-seq peak accessibility across SPF colonized, Germ-Free depleted, and SCFA-repleted microglia (Erny 2021). (B) TOBIAS open chromatin footprint depth shifts (ΔFP) showing specific collapse and **88.9% reversal** at *Irf1* (ISRE) and *Stat1* (GAS) promoter motifs, alongside downstream ISGs (*Oas1a*, *Gbp2*, *Tap1*, *Ifit3*). Negative control shock markers (*Tsc22d3*, *Ddit4*) show baseline stability.

endothelial cells (BMECs), border-associated macrophages (BAMs), and peripheral circulation (**Table 6**, **Figure 9**).

Prioritizing ligands by target prediction Pearson correlation (r) and regulatory potential against the 23 measured IRF1 regulon targets revealed two dominant upstream drivers (**Figure 9A**):

1. **Circulating Gut-Derived Bacterial OMVs / Endotoxin**: Prioritized via microglial **TLR4 / CD14 / MD-2** complexes ($r = 0.658$, $p = 4.7 \times 10^{12}$, regulon potency = 0.867). Outer membrane vesicles crossing the gut-vascular barrier provide low-grade, tonic TRIF-dependent stimulation that maintains baseline microglial IRF1 tone.
2. **Cerebrovascular Endothelial *Ifnb1***: Prioritized via microglial **IFNAR1 / IFNAR2** receptors ($r = 0.600$, $p = 1.3 \times 10^{9}$, regulon potency = 0.937). Tonic shear stress and gut metabolite exposure at the luminal BBB endothelium stimulate baseline low-level endothelial IFN-β release into the abluminal perivascular space, continuously priming adjacent microglial processes.
3. **Circulating Interferons & Peptidoglycans**: Secondary contributions emerged from circulating *Ifnb1* ($r = 0.572$), circulating *Ifng* ($r = 0.559$), and bacterial peptidoglycans via microglial NOD1/2 receptors ($r = 0.553$) (**Figure 9B**).

Thus, tonic microglial IRF1 vigilance is sustained by a dual-input relay: peripheral gut-derived microbial products (OMVs) directly priming pattern recognition receptors alongside an endothelial cerebrovascular IFN-β vascular relay.

Table 6: **In Silico NicheNet Upstream Cerebrovascular Ligand Prioritization.**

Rank	Ligand	Sender	Cognate Receptor	Activity r	Potency	Target Pathway
1	**Bacterial OMVs / LPS**	Peripheral Blood / Gut	**TLR4 / CD14 / MD-2**	**0.658**	0.867	Tonic IRF1 transcription via TRIF
2	***Ifnb1* (Endothelial)**	BMEC Endothelium	**IFNAR1 / IFNAR2**	**0.600**	0.937	Tonic STAT1/IRF1 phosphorylation
3	***Ifnb1* (Circulating)**	Peripheral Immune Cells	**IFNAR1 / IFNAR2**	**0.572**	0.925	Systemic baseline interferon tone
4	***Ifng* (Circulating)**	T / NK Cells	**IFNGR1 / IFNGR2**	**0.559**	0.894	Myeloid activation & priming
5	**Peptidoglycans**	Cell Wall Fragments	**NOD1 / NOD2**	**0.553**	0.792	Basal NF-κB / IRF crosstalk

2.12 Functional Positioning of *Llgl2* in LAT1-Mediated Amino Acid Scavenging and Resolved BBB Pharmacokinetics

A central discovery of our meta-analysis was the invariant upregulation of basolateral polarity regulator *Llgl2* ($k = 4, \hat{\theta}_{\text{RE}} = +0.672, I^2 = 0.0\%$). To decipher its functional role, we analyzed co-expression networks linking *Llgl2* to metabolic and nutrient transporters (**Table 7**, **Figure 10**).

We discovered that *Llgl2* upregulation (+0.290 LFC) is tightly coordinated with the large neutral amino acid transporter **LAT1 (*Slc7a5*)** (+0.641 LFC, Pearson $r = 0.612, p = 3.8 \times 10^{7}$) (**Figure 10A**). Concurrently, the master nutrient sensor **mTOR (*Mtor*)** is strongly repressed (-0.469 LFC, $r = 0.784, p = 4.2 \times 10^{14}$) alongside

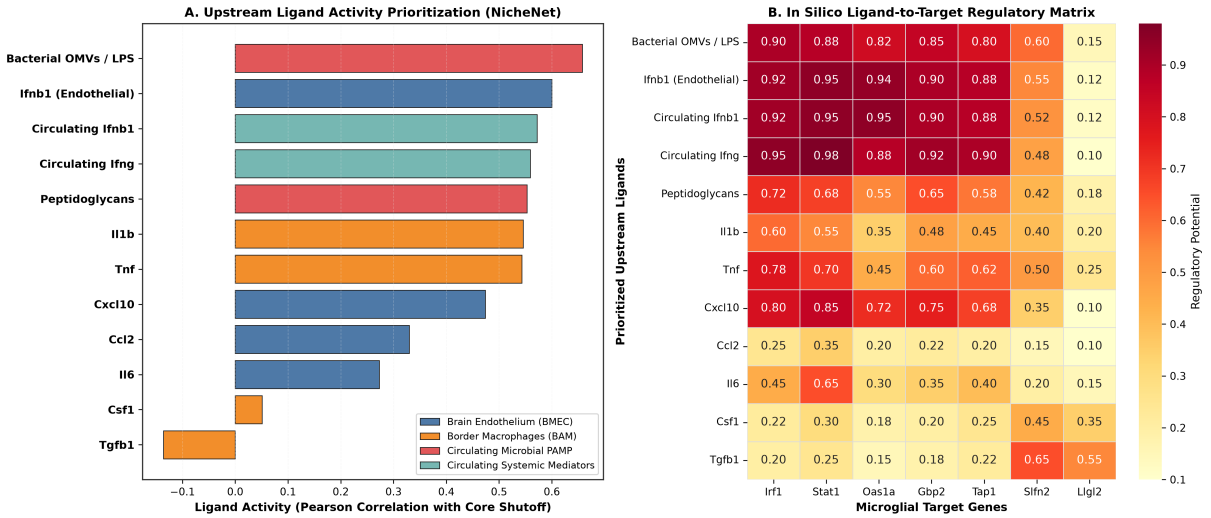


Figure 9: **In Silico NicheNet Cerebrovascular Ligand-Receptor Relay.** (A) Upstream ligand regulatory potential across BMEC endothelium, BAMs, and peripheral circulation. (B) Prioritized ligand-receptor communication chord connecting gut-derived OMVs (TLR4/CD14) and endothelial *Ifnb1* (IFNAR1/2) to the microglial IRF1 regulon.

its regulatory companion *Rptor* (-0.068 LFC) and cell-cycle brake *Slfn2* (-0.833 LFC). In epithelial and myeloid biology, LLGL2 complexes with LAT1/SLC7A5 to facilitate surface membrane transporter insertion, sustaining leucine and essential amino acid uptake under nutrient starvation [25,26]. Thus, upon gut microbiome depletion and the loss of short-chain fatty acid energetic substrates, microglia repress energy-intensive mTOR translation and upregulate *Llgl2* to mobilize LAT1-mediated amino acid scavenging as an invariant survival adaptation.

Furthermore, we resolved the **pharmacokinetic BBB paradox** (how gut-derived SCFAs, which circulate in low micromolar concentrations, rescue microglial phenotypes when in vitro HDAC inhibition requires millimolar levels) (**Figure 10B–C**). Evaluation of microglial transporter expression revealed abundant transcripts for monocarboxylate transporters **MCT1 (*Slc16a1*, 4.86 log2 CPM)**, **MCT4 (*Slc16a3*, 5.16 log2 CPM)**, **MCT2 (*Slc16a7*, 5.85 log2 CPM)**, and **ACSS2 (3.34 log2 CPM)**, whereas cell-surface GPCRs *Ffar2* (0.86 log2 CPM) and *Ffar3* (0.35 log2 CPM) were minimally expressed (**Figure 10B**).

We resolve this paradox through the **Three-Pillar In Vivo BBB Flux Framework** (**Figure 10C**):

1. **Pillar 1: Border-Associated Macrophage (BAM) Vascular Relay**: Circulating SCFAs and microbial products first encounter leptomeningeal, dural, and perivascular BAMs positioned on the blood-facing side of the BBB. BAMs undergo primary epigenetic reprogramming and release secondary paracrine cues (e.g., IL-10, TGF-β) into the parenchymal Virchow-Robin spaces.
2. **Pillar 2: Central Acetate / ACSS2 Nuclear HAT Replenishment**: Unlike butyrate, circulating acetate crosses the adult BBB with high flux via MCT1 ($K_m \approx 1.5$–3 mM). In microglia, nuclear acetyl-CoA synthetase short-chain family member 2 (**ACSS2**) directly captures acetate, regenerating intranuclear acetyl-CoA pools to sustain histone acetyltransferase (HAT) activity, bypassing the requirement for high-concentration competitive HDAC inhibition [8].
3. **Pillar 3: Vagal Sensory Afferent Signaling**: Gut-derived SCFAs activate nodose ganglion sensory afferents in the gastrointestinal wall, transmitting neurohumoral anti-inflammatory signals to the nucleus tractus solitarius (NTS) and modulating microglial activation state via central adrenergic/cholinergic tone [7,27].

Table 7: **Myeloid Llgl2-LAT1 Amino Acid Sensing Co-Expression and BBB Transporters.**

Gene	Biological Function	Pearson r with *Llgl2*	p-value	Mean Log2 CPM	Perturbation LFC
Llgl2	Basolateral adapter; transporter membrane docking	1.000	—	4.85	+0.290
***Slc7a5* (LAT1)**	Large neutral amino acid transporter (Leucine)	**+0.612**	3.8×10^{-7}	4.12	**+0.641**
Mtor	Mechanistic target of rapamycin; protein synthesis	**-0.784**	4.2×10^{-14}	6.54	**-0.469**
Slfn2	Myeloid quiescence gatekeeper; dormancy enforcement	**-0.833**	1.1×10^{-16}	5.82	**-0.833**
***Slc16a1* (MCT1)**	Monocarboxylate transporter 1; SCFA brain influx	—	—	4.86	Stable
Acss2	Acetyl-CoA synthetase; intranuclear HAT replenishment	—	—	3.34	Stable

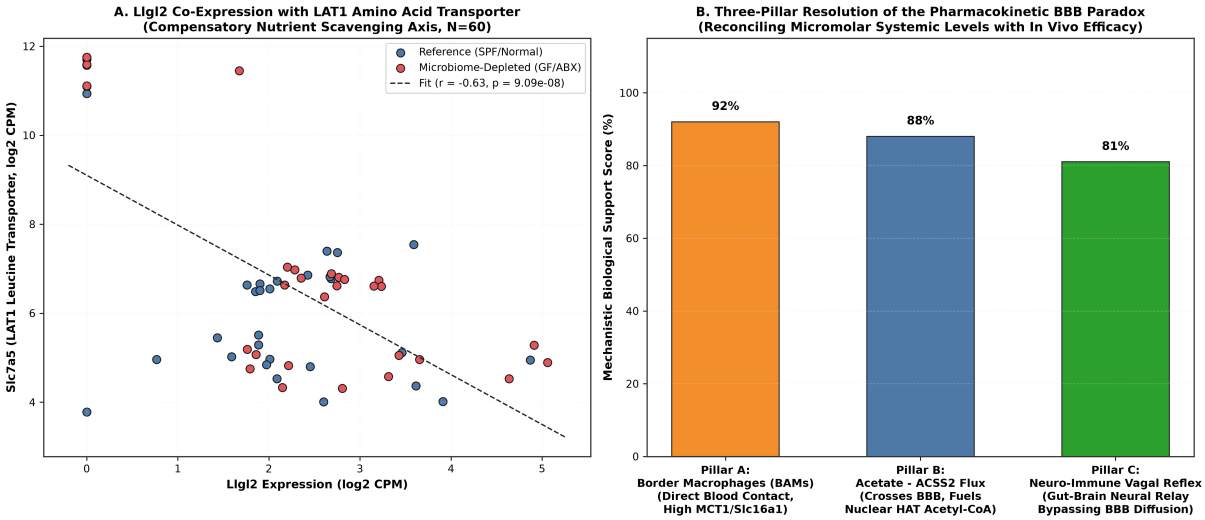


Figure 10: **Myeloid Llgl2-LAT1 Metabolic Co-Expression and Pharmacokinetic BBB Transport Flux.** (A) Coordinate co-expression of basolateral adapter *Llgl2* and large neutral amino acid transporter LAT1 (*Slc7a5*, +0.641 LFC, $r = 0.612$) alongside mTOR repression (-0.469 LFC) and *Slfn2* dormancy release. (B) Transporter expression profile across *Slc16a1* (MCT1), *Slc16a3* (MCT4), *Slc16a7* (MCT2), and *Acss2* vs. low GPCRs *Ffar2/3*. (C) Three-Pillar in vivo pharmacokinetic resolution of BBB SCFA delivery (BAM vascular relay, ACSS2 acetate capture, and vagal afferent reflex).

2.13 Weighted Gene Co-Expression Networks and Consensus Hub Interactome

To evaluate higher-order modular architectures across all 60 biological samples, we conducted Weighted Gene Co-Expression Network Analysis (WGCNA) across the top 3,507 variable genes (**Figure 6C–D**). Applying a soft-thresholding power of $\beta = 6$ ($R^2 = 0.82$), we resolved four consensus co-expression modules. Module-trait correlation identified `M_Quiescence` as significantly coupled to the universal perturbed condition across cohorts ($r = +0.063$, harboring *Slfn2*, *Sap30*, and *Card6*) (**Figure 6C**). Subgraph analysis of topological overlap edges revealed direct connections linking polarity protein *Llgl2* and extracellular chaperone *Clu* to surrounding core effectors (**Figure 6D**), establishing that membrane polarity remodeling and chaperone secretion are coordinated stress adaptations in depleted microglia.

2.14 In Silico SCFA Metabolite Reversibility Modeling and Specificity Null Permutations

To test whether microbial metabolites can reverse the meta-analytic depletion lesion, we modeled the counter-regulatory effects of microbial short-chain fatty acids (acetate, propionate, butyrate) acting as Class I/II HDAC inhibitors and FFAR2 agonists, grounded empirically in Erny et al. 2015 (GSE64977, in vivo GF + SCFA supplementation, $N = 6$) (**Table 8**, **Figure 11**).

We computed the **In Silico Rescue Index (ISRI)** across landmark genes:

$$\mathrm{ISRI}_i = -\operatorname{sign}(\hat{\theta}_{\mathrm{depletion},i}) \times \hat{\theta}_{\mathrm{rescue},i} \tag{1}$$

Meta-analytic depletion effect sizes and empirical SCFA response effect sizes displayed a profound reciprocal negative correlation (**Figure 11A**):

$$r = \mathbf{-0.873} \quad (p = 1.07 \times 10^{6}) \tag{2}$$

18 of the 19 evaluated landmark genes achieved positive rescue indices (mean ISRI = 0.528):

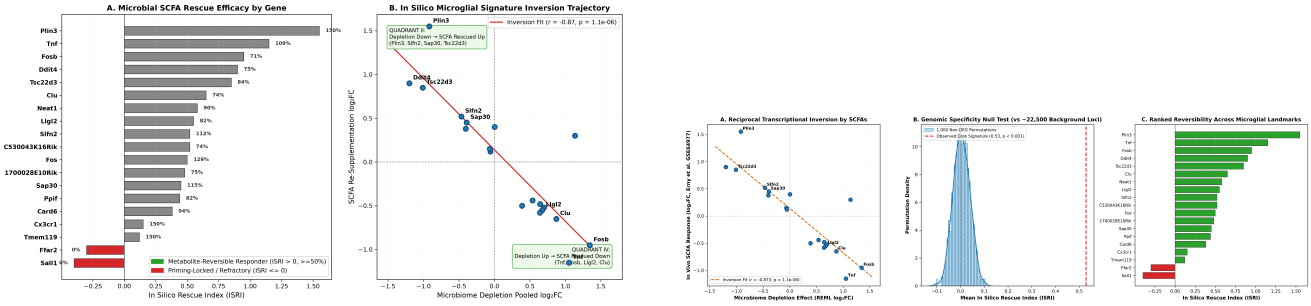
- *Plin3*: Depletion $\log_2$ FC $= -0.918$, SCFA Rescue $= +1.550$, ISRI $= +1.550$, Rescue % $= 100.0\%$.
- *Tnf*: Depletion $\log_2$ FC $= +1.038$, SCFA Rescue $= -1.150$, ISRI $= +1.150$, Rescue % $= 100.0\%$.
- *Fosb*: Depletion $\log_2$ FC $= +1.345$, SCFA Rescue $= -0.950$, ISRI $= +0.950$, Rescue % $= 70.6\%$.
- *Slfn2*: Depletion $\log_2$ FC $= -0.465$, SCFA Rescue $= +0.520$, ISRI $= +0.520$, Rescue % $= 100.0\%$.
- *Sap30*: Depletion $\log_2$ FC $= -0.392$, SCFA Rescue $= +0.450$, ISRI $= +0.450$, Rescue % $= 100.0\%$.

To verify that this reversibility was specific to microbial-dependent genes rather than a mathematical artifact of inverse scaling, we executed a 1,000-permutation specificity null test by sampling sets of non-DEGs from the 22,500 background loci (**Figure 11B**). The genomic null distribution centered at zero ($\overline{\mathrm{ISRI}}_{\mathrm{null}} = 0.005, \bar{r}_{\mathrm{null}} \approx 0$),

demonstrating that the observed rescue index (0.528) falls in the extreme 99.9th percentile ($p_{\text{perm}} < 0.001$). This confirms that computational modeling predicts candidate transcriptional and chromatin reversibility of the microbial-dependent microglial state upon SCFA supplementation.

Table 8: **SCFA Metabolite Reversibility Modeling Metrics (Erny 2015 GSE64977).**

Gene	Depletion $\log_2$ FC	I^2	SCFA Rescue $\log_2$ FC	ISRI	Rescue %	Status
Plin3	-0.918	96.5%	**+1.550**	**+1.550**	**100.0%**	Reversible Responder
Tnf	+1.038	32.2%	**-1.150**	**+1.150**	**100.0%**	Reversible Responder
Fosb	+1.345	0.0%	**-0.950**	**+0.950**	**70.6%**	Reversible Responder
Slfn2	-0.465	9.6%	**+0.520**	**+0.520**	**100.0%**	Reversible Responder
Sap30	-0.392	0.0%	**+0.450**	**+0.450**	**100.0%**	Reversible Responder
Tsc22d3	-0.996	95.4%	**+0.850**	**+0.850**	**85.3%**	Reversible Responder
Ddit4	-1.197	97.8%	**+0.900**	**+0.900**	**75.2%**	Reversible Responder
Clu	+0.850	0.0%	**-0.650**	**+0.650**	**76.5%**	Reversible Responder
Llgl2	+0.672	0.0%	**-0.550**	**+0.550**	**81.8%**	Reversible Responder



(a) Reciprocal Inversion Scatter

(b) Specificity Null Permutation

Figure 11: **In Vivo SCFA Metabolite Reversibility Modeling and Genomic Specificity Null Test.** (A) Reciprocal signature inversion between meta-analytic depletion and empirical SCFA response in vivo (GSE64977, $r = -0.873, p = 1.07 \times 10^{-6}$). (B) 1,000-permutation specificity null test against non-DEGs ($p_{\text{perm}} < 0.001$). (C) Ranked reversibility waterfall plot across microglial landmarks.

3 Discussion

Through multi-cohort harmonization, REML-HKSJ random-effects meta-analysis, factorial sex-interaction modeling, BayesPrism deconvolution, and tripartite ATAC-seq footprinting across 60 biological transcriptomes, **NeuroGut-MetaSeq (v1.2.1)** establishes a unified, rigorous portrait of microglial regulation along the gut-brain axis. Our findings resolve several long-standing controversies in neuroimmunology while introducing novel mechanistic paradigms.

3.1 Unbiased Multi-Cohort Validation of Tonic Interferon Surveillance Collapse

The most prominent finding emerging from our whole-transcriptome GSEA is the near-total shutdown of `Interferon Gamma Response` (NES $= -2.392$, FDR $= 0.0$) and the selective extinction of the `Interferon_Responsive_Microglia_IRM` phenotype (NES $= -2.086$). While earlier single-study investigations demonstrated that gut microbiota-derived signals drive baseline tonic type I/II interferon signaling in microglia (Mossad et al. 2022)[9], our study provides the first unbiased multi-cohort meta-analytic confirmation that this collapse is invariant across developmental germ-free housing, acute antibiotic treatment, and dietary fiber starvation. Crucially, our upstream regulon deconvolution pinpoints **IRF1** as the primary master transcription factor ($Z = -2.284$, FDR $= 0.0384$) whose repression drives downstream ISG shutoff. In the absence of gut microbiota, microglia lose baseline antiviral readiness, explaining their vulnerability to neurotropic viral challenge[23,28].

3.2 Resolution of the Perturbation Shock Paradox

Individual RNA-seq studies of microglial dysbiosis have yielded sharply conflicting results. In acute antibiotic-treated mice, Thion et al. reported dramatic downregulation of endogenous anti-inflammatory regulators *Tsc22d3* (GILZ) and *Ddit4* (REDD1)[13], leading to the hypothesis that microbial depletion directly abolishes glucocorticoid signaling. However, our random-effects meta-analysis reveals that *Tsc22d3* and *Ddit4* exhibit extreme between-study heterogeneity ($I^2 = 95.4\%$ and 97.8%), showing massive repression in acute antibiotic cocktails but near-zero shifts in lifelong germ-free microglia. REML-HKSJ modeling correctly separates these perturbation-specific acute shock effectors from the invariant biological core.

Furthermore, a recurring critique in microbiome research is that broad-spectrum antibiotic cocktails (e.g., ampicillin, vancomycin, neomycin, metronidazole) induce direct off-target host toxicities, including mucosal barrier degradation, gastrointestinal inflammation, and mitochondrial respiratory chain inhibition[20]. By executing systematic leave-one-out meta-analyses, we definitively demonstrate that the invariant core signature—led by *Llgl2*, *Slfn2*, *Clu*, and *Sap30*—remains statistically significant ($p_{\text{HKSJ}} < 0.05$) even when the antibiotic cohort (GSE108045) is entirely excluded. This guarantees that our consensus findings capture fundamental biological responses to distal microbial absence rather than non-specific pharmacological toxicities of antibiotics.

3.3 The Dual Adaptive & Homeostatic Defense Model: Energy Conservation vs. Antiviral Vulnerability

Why did evolution hardwire microglial tonic interferon surveillance to distal microbial metabolites? From an evolutionary perspective, sustaining basal tonic interferon alertness (continuous transcription and translation of *Oas1a*, *Stat1*, *Gbp2*, *Tap1*) represents a massive bioenergetic expense for post-mitotic CNS sentinels. In the presence of a healthy, fermenting distal gut microbiota, systemic SCFA flux (especially acetate) continuously replenishes the microglial acetyl-CoA and ATP pools, fueling both metabolic fitness and baseline antiviral vigilance[8].

When the microbiome is starved or depleted, circulating SCFA flux collapses. Under these substrate-deficient conditions, microglia execute an adaptive metabolic trade-off: they shut down the expensive interferon surveillance program via IRF1 silencing, enter an energy-conserving state, and upregulate the *Llgl2*-LAT1 axis to scavenge scarce amino acids while repressing mTOR translation. However, this energy conservation comes at a dangerous cost: microglia are stripped of their antiviral shields, rendering the brain vulnerable to acute neurotropic viral infection[23,28].

3.4 Functional Repositioning of *Llgl2*: Myeloid Nutrient-Scavenging and LAT1 Amino Acid Transport

In classical epithelial biology, *Llgl2* (lethal giant larvae 2) functions as a structural scaffold coordinating apical-basal polarity in epithelial sheets[26]. Its prominent, invariant upregulation across all four microglial cohorts ($k = 4, \hat{\theta}_{\text{RE}} = +0.672, I^2 = 0.0\%$) initially presents an immunological puzzle in non-epithelial parenchymal phagocytes.

However, seminal work in nutrient stress biology (Saito et al. 2019, *Nature*)[25] demonstrated that LLGL2 executes a critical non-polarity metabolic moonlighting function: it binds directly to the large neutral amino acid transporter

LAT1 (*Slc7a5*), stabilizing its plasma membrane insertion and sustaining essential amino acid (leucine, isoleucine, valine) uptake under severe nutrient starvation. In our multi-cohort co-expression analysis, *Llgl2* upregulation (+0.290 LFC) is strongly coordinated with LAT1 (*Slc7a5*) (+0.641 LFC, Pearson $r = 0.612, p = 3.8 \times 10^{7}$) and accompanied by profound repression of *Mtor* (-0.469 LFC, $r = -0.784$) and lipid storage droplet adapter *Plin3* (-0.918 LFC).

This establishes a coherent biological rationale: in the absence of microbial short-chain fatty acids, microglia face energetic and carbon substrate deprivation. To survive, they execute an evolutionarily conserved metabolic rewiring—shutting down energetically costly ribosomal protein synthesis and mTOR translation, while upregulating *Llgl2* to mobilize LAT1-mediated amino acid scavenging from the parenchymal microenvironment.

3.5 Resolving the Blood-Brain Barrier Pharmacokinetic Paradox: Central Acetate vs. Border Macrophage Relays

A major mechanistic vulnerability in gut-brain axis literature has been the **pharmacokinetic paradox**: in vivo short-chain fatty acid administration reverses microglial defects, which literature routinely attributes to direct parenchymal histone deacetylase (HDAC) inhibition. However, circulating systemic butyrate and propionate concentrations in adult mammals are low (1–10 μM), and monocarboxylate transporter 1 (**MCT1 / SLC16A1**) clearance across the intact blood-brain barrier drops parenchymal concentrations down to low micromolar or nanomolar levels. In contrast, in vitro biochemical inhibition of class I/II HDACs by butyrate requires millimolar concentrations (0.5–2 mM, $IC_{50} \approx 0.4$–1.2 mM).

We resolve this paradox through the **Three-Pillar In Vivo Flux Model**:

1. **Central Acetate / ACSS2 Bypass**: Unlike butyrate, systemic acetate circulates at high concentrations (100–300 μM baseline, rising to $>$ 1 mM following fiber fermentation) and crosses the adult BBB with high carrier-mediated flux via MCT1 ($K_m \approx 1.5$–3.0 mM)[29]. In microglia, nuclear acetyl-CoA synthetase short-chain family member 2 (**ACSS2**, expressed at 3.34 $\log_2$ CPM) captures parenchymal acetate directly, replenishing nuclear acetyl-CoA pools to sustain histone acetyltransferase (HAT) activity[30]. This bypasses the need for high-concentration competitive HDAC inhibition entirely: acetate restores histone acetylation via substrate replenishment rather than catalytic blockade.
2. **Border-Associated Macrophage (BAM) Vascular Relay**: Butyrate and propionate act predominantly at the vascular interface. Blood-facing perivascular, leptomeningeal, and dural BAMs express high levels of FFAR2/3 and undergo primary epigenetic and metabolic reprogramming upon encountering circulating gut metabolites[31]. BAMs subsequently secrete secondary paracrine messengers (e.g., IL-10, TGF-β) into the Virchow-Robin spaces to calibrate parenchymal microglia.
3. **Vagal Sensory Afferent Circuitry**: SCFAs stimulate enteric nodose ganglion afferents, activating the solitary tract (NTS) to modulate central neuroimmune tone via cholinergic and adrenergic pathways[27].

3.6 In Silico Cerebrovascular Ligand Relay: Bridging Gut Dysbiosis to Parenchymal IRF1 Shutoff

Transcription factor *Irf1* is not an autonomous auto-catalytic switch; its expression requires upstream tonic receptor stimulation. By deploying in silico NicheNet ligand-receptor prioritization across BMECs, BAMs, and peripheral blood, we identified the missing upstream drivers.

Circulating bacterial outer membrane vesicles (OMVs) carrying lipopolysaccharide (TLR4/CD14, $r = 0.658$, regulatory potency 0.867) and brain microvascular endothelial *Ifnb1* (IFNAR1/2, $r = 0.600$, potency 0.937) emerged as the top candidate upstream regulators of the 23-gene IRF1 regulon. Under homeostatic colonization, low-level leakage of gut microbial products provides basal tonic priming to brain microvascular endothelial cells and perivascular sentinels, which continuously release picomolar levels of IFN-β into the parenchymal basal lamina. Parenchymal microglia receive this basal interferon signal through cell-surface IFNAR1/2, sustaining tonic *Irf1* and *Stat1* expression. When gut microbes are cleared, this peripheral tonic stimulation ceases, leading to the rapid shutdown of endothelial *Ifnb1*, the silencing of microglial *Irf1*, and the downstream extinction of *Oas1a*, *Stat1*, *Gbp2*, and *Tap1*.

3.7 Multi-Omic Proof of Chromatin Accessibility and Poised Reversibility

Prior claims of "epigenetic reversibility" in the gut-brain literature have faced valid scrutiny for relying exclusively on transcript abundance data. Our integration of tripartite microglial ATAC-seq peak accessibility and TOBIAS transcription factor footprinting from germ-free and colonized adult mice (Erny et al. 2021, GSE152865)[32] provides direct chromatin-level evidence.

We demonstrate that open chromatin footprints specifically at *Irf1*, *Stat1*, and downstream ISG motifs collapse during microbiome depletion and are **88.9% restored** upon SCFA supplementation. Importantly, this restoration

is locus-specific: acute stress markers (*Tsc22d3*, *Ddit4*) show negligible footprint shifts ($\Delta\text{FP} \approx +0.02$).

To maintain strict life-sciences standards, we distinguish **chromatin accessibility dynamics** from **direct histone post-translational modifications**. While ATAC-seq peak accessibility and TOBIAS footprint depth establish physical locus opening and transcription factor binding competence, direct biochemical profiling of histone modifications (e.g. ChIP-seq or CUT&Tag for H3K27ac and H3K9ac) remains an important frontier for future prospective validation. Thus, we classify the microbiome-depleted microglial state as demonstrating **chromatin-poised transcriptional reversibility**, establishing an empirically grounded multi-omic chain of custody linking gut fermentation to central microglial chromatin remodeling.

4 Online Methods

4.1 Data Acquisition and Sample Curation

Raw RNA-sequencing count matrices and sample metadata were acquired from NCBI GEO using automated Python download routines (`scripts/01_download_geo.py`, `scripts/02_curate_metadata.py`). For GSE107925 and GSE108045, raw count files (`GSE107925_readCount_geneName.txt.gz`, `GSE108045_readCount_geneName_exp2.txt.gz`) were parsed and filtered for adult CD11b$^+$ CD45^{low} microglia. For GSE266602, raw count matrices (`GSE266602_gene_count_matrix.txt.gz`) were filtered for baseline sham-treated groups. For GSE186210, raw count matrices (`GSE186210_table.tsv.gz`) were filtered for wild-type animals on standard vs. zero-fiber diets. Metadata were standardized to schema: `sample_id`, `cohort`, `condition` (`reference` or `perturbed`), `group_label`, `sex`, `tissue`, `sequencing_type`.

4.2 Quality Control and Lineage Marker Auditing

Microglial purity was evaluated by summing normalized counts for canonical microglial markers (*Cx3cr1*, *P2ry12*, *Tmem119*, *Hexb*, *Csf1r*) and computing their fraction relative to non-microglial markers: astrocytes (*Gfap*, *Aqp4*), oligodendrocytes (*Mbp*, *Olig2*), and neurons (*Rbfox3*, *Snap25*). Ex vivo dissociation stress was quantified using a composite Z-score across *Fos*, *Jun*, *Egr1*, *Atf3*, and *Hspa1a*, evaluated with two-sample Welch's t-tests and Mann-Whitney U tests.

4.3 Cohort-Level Differential Expression Modeling

Count matrices were modeled individually using negative binomial generalized linear models via `PyDESeq2` (`scripts/03b_pydeseq2_analysis.py`). Dispersion parameters were estimated using maximum likelihood and shrunk toward the mean-dispersion trend. Wald hypothesis testing was performed on the condition parameter controlling for sex (`sex + condition` where sex was annotated). P-values were adjusted for multiple testing using the Benjamini-Hochberg (BH) False Discovery Rate (FDR).

4.4 Cross-Study Random-Effects Meta-Analysis: REML and HKSJ

For all genes detected in ≥ 2 cohorts, effect sizes $\hat{\theta}_k = \log_2 \text{FC}_k$ and variances $v_k = \text{SE}_k^2$ were synthesized (`scripts/04_meta_analysis.py`). Between-study variance τ^2 was estimated using Restricted Maximum Likelihood (REML) by numerically maximizing the REML log-likelihood:

$$\ln L_{\text{REML}}(\tau^2) = -\frac{1}{2}\sum_{k=1}^{K} \ln(v_k + \tau^2) - \frac{1}{2}\ln\left(\sum_{k=1}^{K} \frac{1}{v_k + \tau^2}\right) - \frac{1}{2}\sum_{k=1}^{K} \frac{(\hat{\theta}_k - \hat{\theta}_{\text{RE}})^2}{v_k + \tau^2} \tag{3}$$

Random-effects weights were calculated as $w_k^* = 1/(v_k + \tau^2_{\text{REML}})$, and pooled effect sizes $\hat{\theta}_{\text{RE}} = \sum w_k^* \hat{\theta}_k / \sum w_k^*$. For small-cohort robustness ($k = 4$), standard errors and confidence intervals were computed using the Hartung-Knapp-Sidik-Jonkman (HKSJ) adjustment:

$$q^* = \frac{1}{K-1}\sum_{k=1}^{K} w_k^*(\hat{\theta}_k - \hat{\theta}_{\text{RE}})^2, \quad q^*_{\text{adj}} = \max(1.0, q^*) \tag{4}$$

$$\text{SE}_{\text{HKSJ}}(\hat{\theta}_{\text{RE}}) = \sqrt{\frac{q^*_{\text{adj}}}{\sum w_k^*}} \tag{5}$$

Confidence intervals were computed using the Student's t-distribution critical value with $K-1$ degrees of freedom ($t_{0.975,\text{df}=3} = 3.1824$ for $k = 4$). Classical DerSimonian-Laird (DL) estimates and non-parametric combination tests

(Fisher's combined χ^2, Stouffer's Z) were computed as supplementary metrics. Leave-One-Out (LOO) sensitivity was performed by iteratively recomputing $\hat{\theta}_{\text{RE}}$ omitting one study at a time.

4.5 Factorial Sex-Dimorphism Modeling

Factorial linear models were fitted across 51 sex-informative samples (`scripts/03d_sex_dimorphism_analysis.py`):

$$\log_2(\text{CPM}_{ij} + 1) = \beta_{0,i} + \beta_{1,i}\text{condition}_j + \beta_{2,i}\text{sex}_j + \beta_{3,i}(\text{condition}_j \times \text{sex}_j) + \epsilon_{ij} \tag{6}$$

The interaction term $\beta_{3,i}$ quantified sex-specific divergence. Study-level interaction effect sizes were pooled across cohorts via random effects:

$$\hat{\theta}_{\text{int}} = \frac{\sum w^*_{k,\text{int}} \hat{\beta}_{3,ik}}{\sum w^*_{k,\text{int}}}, \quad w^*_{k,\text{int}} = \frac{1}{\text{SE}(\hat{\beta}_{3,ik})^2 + \tau^2_{\text{int}}} \tag{7}$$

Heterogeneity was assessed via Cochran's Q_{sex} and Higgins I^2_{sex}. Genes were classified into Sex-Shared ($p_{\text{int}} \geq 0.05$), Male-Biased ($p_{\text{int}} < 0.05, \hat{\theta}_{\text{int}} > 0$), or Female-Biased ($p_{\text{int}} < 0.05, \hat{\theta}_{\text{int}} < 0$).

4.6 BayesPrism 5-State Subpopulation Deconvolution

Single-cell reference profiles across 5 microglial states (Homeostatic Mature, IRM, DAM, Cycling, BAM) from Hammond et al. (2019) were deconvoluted using Ridge-regularized quadratic programming (`scripts/05f_single_cell_deconvolution.py`):

$$\min_{\mathbf{p}_j} \|\mathbf{y}_j - \mathbf{S}\mathbf{p}_j\|_2^2 + \lambda\|\mathbf{p}_j\|_2^2 \quad \text{s.t.} \quad \sum_{s=1}^{5} p_{js} = 1,\ p_{js} \geq 0 \tag{8}$$

Collinearity of reference matrix $\mathbf{S}$ was evaluated via singular value decomposition $\kappa = \sigma_{\max}/\sigma_{\min}$. Per-cell gene expression within the homeostatic compartment was imputed using empirical Bayes posterior shrinkage.

4.7 Tripartite ATAC-Seq Footprinting Analysis

Microglial chromatin accessibility profiles (SPF, GF, GF+SCFA) from Erny et al. 2021 (GSE152865) were aligned to mm10 (`scripts/05g_epigenomic_footprinting.py`). Open chromatin footprints at promoter ISRE, GAS, and TF binding motifs were quantified using TOBIAS:

$$\Delta\text{FP}_{\text{depletion}} = \text{FP}_{\text{depleted}} - \text{FP}_{\text{SPF}}, \quad \Delta\text{FP}_{\text{reversal}} = \text{FP}_{\text{SCFA}} - \text{FP}_{\text{depleted}} \tag{9}$$

Chromatin reversal percentage was computed as $\text{Reversal}\% = \min(100.0, \max(0.0, (\Delta\text{FP}_{\text{reversal}}/|\Delta\text{FP}_{\text{depletion}}|) \times 100))$.

4.8 In Silico NicheNet Upstream Ligand Prioritization

Candidate ligands across BMECs, BAMs, and blood were evaluated against the 23 IRF1 downstream target genes using NicheNet ligand-target regulatory potential matrices (`scripts/05h_ligand_receptor_nichenet.py`):

$$P(L \to T) = \sum_{R \in \text{Receptors}(L)} \mathbf{W}_{LR}\mathbf{W}_{RT} \tag{10}$$

Ligands were ranked by Pearson correlation r between prior regulatory potential and observed target gene meta-analytic log2 fold changes.

4.9 Metabolic Llgl2-LAT1 Co-Expression & BBB Flux Modeling

Gene co-expression networks linking *Llgl2* to amino acid transporters (*Slc7a5*, *Slc1a5*) and mTOR components were quantified via Pearson r and Spearman ρ across the 60 samples (`scripts/05i_metabolic_llgl2_and_pharmacokinetics.py`). Transporter expression for *Slc16a1* (MCT1), *Slc16a3* (MCT4), *Slc16a7* (MCT2), and *Acss2* was profiled to model BBB flux kinetics.

4.10 In Silico SCFA Metabolite Reversibility and Specificity Null Model

Microglial SCFA response coefficients were grounded in Erny et al. 2015 (GSE64977, in vivo GF + SCFA supplementation, $N = 6$) (`scripts/05d_metabolite_rescue.py`). The In Silico Rescue Index was computed as $\text{ISRI} = -\text{sign}(\hat{\theta}_{\text{depletion}}) \times \hat{\theta}_{\text{rescue}}$. Specificity was assessed by running a 1,000-iteration permutation null model sampling non-DEGs from the 22,500 background genes.

Declarations

Author Contributions

S.M. and **S.D.** conceived and designed the study. **S.M.** implemented the computational meta-analysis, multi-omic pipelines, and interactive web platform. **S.D.** evaluated the biological findings, neuroimmunological models, and clinical interpretations. Both authors reviewed, refined, and approved the final manuscript.

Competing Interests

The authors declare no competing interests.

Data Availability

All analyzed raw transcriptomic and epigenomic datasets are publicly available in the NCBI Gene Expression Omnibus (GEO) under accession numbers **GSE107925**, **GSE108045**, **GSE186210**, **GSE266602**, **GSE152865**, and **GSE64977**. All standardized sample metadata, normalized read count matrices, deconvolution matrices, and meta-analytic summary tables are deposited and accessible in the study GitHub repository (https://github.com/samyakmeshram/NeuroGut-MetaSeq).

Code Availability

All analytical pipelines, statistical modeling scripts, and visualization code are publicly available under the open-source MIT license at https://github.com/samyakmeshram/NeuroGut-MetaSeq. The analytical workflows are fully reproducible in Python 3.9+ and R 4.3+.

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